

Chapter Seven

Numerical Differentiation and Integration

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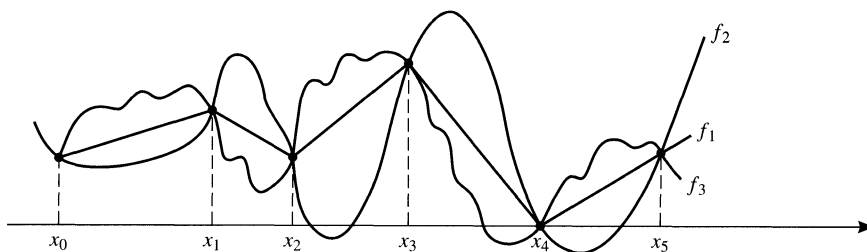
7.1 Numerical Differentiation and Richardson Extrapolation

If the values of a function f are given at a few points, say $x_0, x_1, \dots, x_n$, can that information be used to estimate a derivative $f'(c)$ or an integral $\int_a^b f(x) dx$? The answer is a qualified *yes*.

Let us begin by observing that from the values $f(x_0), f(x_1), \dots, f(x_n)$ alone it is impossible to infer very much about f unless we are informed also that f belongs to some relatively small family of functions. Thus, if f is allowed to range over the family of all continuous real-valued functions, the values $f(x_i)$ are almost useless. Figure 7.1 illustrates three continuous functions that take the same values at six points.

On the other hand, if we know that f is a polynomial of degree at most n , the values at $n + 1$ points completely determine f , by the theory of interpolation in Section 6.1 (p. 308). In this case, we recover f precisely and can then compute $f'(c)$ or $\int_a^b f(x) dx$ with complete confidence. In most realistic situations, however,

FIGURE 7.1
Three continuous
functions through
six points



the information at hand will not fully determine f , and any numerical estimate of its derivative or integral should be viewed skeptically unless it is accompanied by some bound on the errors involved.

Numerical Differentiation

We illustrate these matters by examining a formula for numerical differentiation that emerges directly from the limit definition of $f'(x)$:

$$f'(x) \approx \frac{1}{h} [f(x + h) - f(x)] \tag{1}$$

For a linear function, $f(x) = ax + b$, the approximate Formula (1) is *exact*; that is, it yields the correct value of $f'(x)$ for any nonzero value of h . The formula may be exact in other cases too, but only fortuitously. Let us therefore attempt to assess the error involved in this formula for numerical differentiation. The starting point is Taylor’s Theorem in this form:

$$f(x + h) = f(x) + hf'(x) + \frac{h^2}{2} f''(\xi) \tag{2}$$

Here ξ is a point in the open interval between x and $x + h$. For the validity of Equation (2), f and f' should be continuous on the closed interval between x and $x + h$, and f'' should exist on the corresponding open interval. A rearrangement of Equation (2) yields

$$f'(x) = \frac{1}{h} [f(x + h) - f(x)] - \frac{h}{2} f''(\xi) \tag{3}$$

Equation (3) is more useful than Equation (1) because now on a large class of functions as described above, an error term is available along with the basic numerical formula. Notice that the error term in Equation (3) has two parts: a power of h and a factor involving some higher-order derivative of f . The latter gives us an indication of the class of functions to which the error estimate is applicable. The h -term in the error makes the entire expression converge to 0 as h approaches 0. The rapidity of this convergence will depend on the power of h . These remarks apply to many error estimates in numerical analysis: There will usually be a power of h and a factor telling us to what smoothness class the function must belong so that the estimate is valid.

EXAMPLE 1 If Formula (1) is used to compute the derivative of $f(x) = \cos x$ at $x = \pi/4$ with $h = 0.01$, what is the answer and how accurate is it?

Solution Using a calculator, we find

$$\begin{aligned} f'(x) &\approx \frac{1}{h} [f(x + h) - f(x)] = \frac{1}{0.01} [0.70000\,0476 - 0.70710\,6781] \\ &= -0.71063\,051 \end{aligned}$$

The error term in Equation (3) can be estimated like this:

$$\left| \frac{h}{2} f''(\xi) \right| = 0.005 |\cos \xi| \leq 0.005$$

We can obtain a sharper bound by using the fact that $\pi/4 < \xi < \pi/4 + h$, so that $|\cos \xi| < 0.707107$. This gives the bound 0.0035355. The actual error is

$$-\sin \frac{\pi}{4} + 0.71063051 = 0.003523729 \quad \blacksquare$$

The term $-(h/2)f''(\xi)$ in Equation (3) is called the **truncation error**. It is the error that arises because, at some stage in the derivation, a Taylor series has been truncated. In this case, the approximate Formula (1) was obtained by truncating the series

$$f(x + h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + \frac{h^3}{3!} f'''(x) + \cdots$$

as follows:

$$f(x + h) \approx f(x) + hf'(x)$$

As we shall see presently, truncation error and roundoff error play equally important roles in the use of Formula (1) and others like it.

A glance at Equation (3) shows that to compute $f'(x)$ accurately, the step size h must be small. Therefore, let us perform an experiment in which h converges to 0 through a sequence of values, and the corresponding approximations to $f'(x)$ are computed. We select $f(x) = \tan^{-1} x$ and use the point $x = \sqrt{2}$. The result should be $f'(x) = (x^2 + 1)^{-1}$ at $\sqrt{2}$, which is $1/3$. Here is an algorithm for this task:

```
f(x) := tan-1 x
input s ← √2; h ← 1; M ← 26
F1 ← f(s)
for k = 0 to M do
    F2 ← f(s + h)
    d ← F2 - F1
    r ← d/h
    output k, h, F2, F1, d, r
    h ← h/2
end do
```

Some output from a 32-bit computer is shown here:

k	h	F ₂	F ₁	d	r
4	0.62 × 10 ⁻¹	0.97555 095	0.95531 660	0.02023 435	0.32374 954
12	0.24 × 10 ⁻³	0.95539 796	0.95531 660	0.00008 136	0.33325 195
20	0.95 × 10 ⁻⁶	0.95531 690	0.95531 660	0.00000 030	0.31250 000
24	0.60 × 10 ⁻⁷	0.95531 666	0.95531 660	0.00000 006	1.00000 000
26	0.15 × 10 ⁻⁷	0.95531 660	0.95531 660	0.00000 000	0.00000 000

In each line, d is computed as the difference $F_2 - F_1$, and r is computed as the ratio d/h . Because of **subtractive cancellation**, d has progressively fewer significant digits, until finally $d = 0$ and $r = 0$. The best value of r is obtained when $k = 12$, and it has four correct digits, if rounded to four decimal places. At this value of k , we notice that d has four significant digits. As k increases, the number of significant digits in d decreases, and of course r cannot have any more significant digits than d . Thus, roundoff error prevents us from getting accurate values when h is small. To have more precision in d , we would require more precision in F_1 and F_2 , and that would require higher precision in the basic calculations. We can use multiple precision or carry out the calculations on a machine with a longer word length. The step at which the answers become worse will depend on the word length (or more exactly, the unit roundoff error) in the particular machine used.

Numerical differentiation formulas find their most important application in the numerical solution of differential equations. A common stratagem is to replace derivatives by approximations, such as the one given in Equation (1). The precision of such numerical differentiation formulas is often judged simply by the power of h in the error term. The higher the power of h the better because h is always a small number. In this assessment, Formula (2) fares poorly, as the error is $\mathcal{O}(h)$. A superior formula is

$$f'(x) \approx \frac{1}{2h} [f(x+h) - f(x-h)] \quad (4)$$

This is derived from two cases of Taylor's Theorem:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + \frac{h^3}{3!} f'''(\xi_1) \quad (5)$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(x) - \frac{h^3}{3!} f'''(\xi_2) \quad (6)$$

Subtracting one of these equations from the other and rearranging, we obtain

$$f'(x) = \frac{1}{2h} [f(x+h) - f(x-h)] - \frac{h^2}{12} [f'''(\xi_1) + f'''(\xi_2)] \quad (7)$$

This is a more favorable result because of the h^2 term in the error. Notice, however, the presence of f''' in the error. This error term is applicable if f''' exists.

The error term in Equation (7) can be simplified if we make the slight additional assumption that the function f''' is continuous on $[x-h, x+h]$. Let M and m denote the greatest and least values of f''' on the interval $[x-h, x+h]$. Then $f'''(\xi_1)$, $f'''(\xi_2)$, and $c \equiv [f'''(\xi_1) + f'''(\xi_2)]/2$ all lie in the interval $[m, M]$. Because f''' is continuous, it assumes the value c at some point ξ in $[x-h, x+h]$. Hence,

$$f'''(\xi) = \frac{1}{2} [f'''(\xi_1) + f'''(\xi_2)]$$

When this expression is substituted in Equation (7), the result is

$$f'(x) = \frac{1}{2h} [f(x+h) - f(x-h)] - \frac{h^2}{6} f'''(\xi) \quad (8)$$

An important formula for second derivatives is obtained by extending Equations (5) and (6) by one more term and then adding the equations. After rearrangement and applying the method used previously, we have

$$f''(x) = \frac{1}{h^2} [f(x+h) - 2f(x) + f(x-h)] - \frac{h^2}{12} f^{(4)}(\xi) \tag{9}$$

for some $\xi \in (x-h, x+h)$. This formula is often used in the numerical solution of second-order differential equations.

EXAMPLE 2 Use a computer to approximate $f'(x)$, where $f(x) = \tan^{-1} x$ and $x = \sqrt{2}$. Use Equation (8) with step size h approaching 0. Recall that the correct value is $1/3$.

Solution A suitable algorithm follows:

```
f(x) := tan-1 x
input s ← √2; h ← 1; M ← 26
for k = 0 to M do
    F2 ← f(s + h)
    F1 ← f(s - h)
    d ← F2 - F1
    r ← d/(2h)
    output k, h, F2, F1, d, r
    h ← h/2
end do
```

Some of the output from a 32-bit machine is shown here:

<i>k</i>	<i>h</i>	<i>F</i> ₂	<i>F</i> ₁	<i>d</i>	<i>r</i>
2	0.25	1.02972 674	0.86112 982	0.16859 692	0.33719 385
10	0.9765 × 10 ⁻³	0.95564 199	0.95499 092	0.00065 106	0.33334 351
18	0.3815 × 10 ⁻⁵	0.95531 786	0.95531 535	0.00000 250	0.32812 500
26	0.1490 × 10 ⁻⁷	0.95531 660	0.95531 660	0.00000 000	0.00000 000

Again, we see that because of **subtractive cancellation**, there is a pronounced *deterioration* in accuracy as h approaches 0. The output begins to exhibit this phenomenon at $k = 9$. The values of F_1 and F_2 are very close to each other, and their difference, d , suffers a severe loss of significant digits. Eventually, as $h \rightarrow 0^+$, the values of F_1 and F_2 will be identical in the machine, and a value 0 will be computed as the derivative. This occurs at $k = 26$. This behavior will appear at different k values on computers with different word lengths. ■

The numerical differentiation of functions known only empirically is a risky procedure because errors in the data will be magnified by the process. This is quite easily seen from Equation (8), for example. If the *sampling points* $x \pm h$ are accurately determined whereas the ordinates $f(x \pm h)$ are inaccurate, the errors in the ordinates will be multiplied by $1/(2h)$. Since h will be small, the influence of the errors will be large. (This phenomenon is *not* present in numerical integration.) Thus, numerical differentiation of *empirical* data should be avoided or undertaken with great caution.

Differentiation via Polynomial Interpolation

A general approach to numerical differentiation and integration can be based on polynomial interpolation. Suppose that we have $n + 1$ values of a function f at points $x_0, x_1, \dots, x_n$. A polynomial that interpolates f at the nodes x_i can be written in the Lagrange form of Equation (9) in Section 6.1 (p. 312). Let us include the error term from Theorem 2 in that section, obtaining

$$f(x) = \sum_{i=0}^n f(x_i) \ell_i(x) + \frac{1}{(n+1)!} f^{(n+1)}(\xi_x) w(x) \quad (10)$$

Here we have written $w(x) = \prod_{i=0}^n (x - x_i)$. Taking derivatives in Equation (10), we have

$$f'(x) = \sum_{i=0}^n f(x_i) \ell'_i(x) + \frac{1}{(n+1)!} f^{(n+1)}(\xi_x) w'(x) + \frac{1}{(n+1)!} w(x) \frac{d}{dx} f^{(n+1)}(\xi_x)$$

If x is one of the nodes, say $x = x_\alpha$, then the preceding equation simplifies, because $w(x_\alpha) = 0$, and the result is

$$f'(x_\alpha) = \sum_{i=0}^n f(x_i) \ell'_i(x_\alpha) + \frac{1}{(n+1)!} f^{(n+1)}(\xi_{x_\alpha}) w'(x_\alpha)$$

This, in turn, can be simplified by computing $w'(x_\alpha)$. To do this, we note that

$$w'(x) = \sum_{i=0}^n \prod_{\substack{j=0 \\ j \neq i}}^n (x - x_j) \quad \text{so} \quad w'(x_\alpha) = \prod_{\substack{j=0 \\ j \neq \alpha}}^n (x_\alpha - x_j)$$

The final differentiation formula with error term is

$$f'(x_\alpha) = \sum_{i=0}^n f(x_i) \ell'_i(x_\alpha) + \frac{1}{(n+1)!} f^{(n+1)}(\xi_{x_\alpha}) \prod_{\substack{j=0 \\ j \neq \alpha}}^n (x_\alpha - x_j) \quad (11)$$

This formula is particularly well suited for nonequally spaced nodes.

EXAMPLE 3 Give the explicit form of Equation (11) when $n = 2$ and $\alpha = 1$.

Solution The three cardinal functions for Lagrange interpolation in this case are

$$\begin{aligned} \ell_0(x) &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} \\ \ell_1(x) &= \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} \\ \ell_2(x) &= \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} \end{aligned}$$

Their derivatives are

$$\ell'_0(x) = \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}$$

$$\ell'_1(x) = \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}$$

$$\ell'_2(x) = \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)}$$

Evaluating at x_1 , we obtain

$$\ell'_0(x_1) = \frac{x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)}$$

$$\ell'_1(x_1) = \frac{2x_1 - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)}$$

$$\ell'_2(x_1) = \frac{x_1 - x_0}{(x_2 - x_0)(x_2 - x_1)}$$

The numerical differentiation formula with its error term is thus

$$\begin{aligned} f'(x_1) &= f(x_0) \frac{x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} \\ &\quad + f(x_1) \frac{2x_1 - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \\ &\quad + f(x_2) \frac{x_1 - x_0}{(x_2 - x_0)(x_2 - x_1)} \\ &\quad + \frac{1}{6} f'''(\xi_{x_1})(x_1 - x_0)(x_1 - x_2) \end{aligned} \quad (12)$$

EXAMPLE 4 In Example 3, what formula results if the nodes are equally spaced?

Solution Let $x_0 = x_1 - h$ and $x_2 = x_1 + h$. Then from Equation (12) we obtain (with $x = x_1$)

$$f'(x) = f(x - h) \left(\frac{-1}{2h} \right) + f(x + h) \left(\frac{1}{2h} \right) - \frac{1}{6} f'''(\xi_x) h^2$$

This is identical to Equation (8), which was previously derived. ■

Richardson Extrapolation

We shall now indicate how a procedure known as **Richardson extrapolation** can be used to coax more accuracy out of some numerical formulas. Consider Equations (5) and (6), extended with higher-order terms. We assume that $f(x)$ is represented by its Taylor series

$$f(x + h) = \sum_{k=0}^{\infty} \frac{1}{k!} h^k f^{(k)}(x) \quad (13)$$

$$f(x - h) = \sum_{k=0}^{\infty} \frac{1}{k!} (-1)^k h^k f^{(k)}(x) \quad (14)$$

If the second equation is subtracted from the first, all terms with an even value of k will cancel, leaving

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{2}{3!}h^3f'''(x) + \frac{2}{5!}h^5f^{(5)}(x) + \dots$$

A rearrangement yields

$$f'(x) = \frac{1}{2h}[f(x+h) - f(x-h)] - \left[\frac{1}{3!}h^2f^{(3)}(x) + \frac{1}{5!}h^4f^{(5)}(x) + \frac{1}{7!}h^6f^{(7)}(x) + \dots \right]$$

This equation has the form

$$L = \varphi(h) + a_2h^2 + a_4h^4 + a_6h^6 + \dots \tag{15}$$

where L stands for $f'(x)$ and $\varphi(h)$ stands for the numerical differentiation formula (4); that is,

$$\varphi(h) = \frac{1}{2h}[f(x+h) - f(x-h)]$$

where x is assigned a numerical value, say s , at which the derivative is sought. The numerical procedure is designed to estimate L . The function φ is such that $\varphi(h)$ can be evaluated for $h > 0$, but it cannot be evaluated for $h = 0$. Thus, we can only let h approach 0 in our attempt to determine L . For each $h > 0$, the error is given by the series of terms $a_2h^2 + a_4h^4 + \dots$. Assuming $a_2 \neq 0$, we see that the first term, a_2h^2 , is greater than the others when h is sufficiently small. We shall therefore look for a way of eliminating this dominant term, a_2h^2 . Our analysis depends only on Equation (15) and will be applicable to other numerical processes; in particular, it is used in Section 7.4 (p. 502) with the trapezoid rule.

Write out Equation (15) with h replaced by $h/2$:

$$L = \varphi(h/2) + a_2h^2/4 + a_4h^4/16 + a_6h^6/64 + \dots \tag{16}$$

The leading term in the error series, a_2h^2 , can be eliminated between Equations (15) and (16) by multiplying the latter by 4 and subtracting the former from it. Here is the work:

$$\begin{array}{rcl} L & = & \varphi(h) + a_2h^2 + a_4h^4 + a_6h^6 + \dots \\ 4L & = & 4\varphi(h/2) + a_2h^2 + a_4h^4/4 + a_6h^6/16 + \dots \\ \hline 3L & = & 4\varphi(h/2) - \varphi(h) - 3a_4h^4/4 - 5a_6h^6/16 - \dots \end{array}$$

Thus, we have

$$L = \frac{4}{3}\varphi(h/2) - \frac{1}{3}\varphi(h) - a_4h^4/4 - 5a_6h^6/16 - \dots \tag{17}$$

Equation (17) embodies the first step in Richardson extrapolation. It shows that a simple combination of $\varphi(h)$ and $\varphi(h/2)$ furnishes an estimate of L with accuracy $\mathcal{O}(h^4)$.

EXAMPLE 5 Recompute the derivative in Example 2, incorporating Richardson extrapolation.

Solution A suitable algorithm is as follows:

```
f(x) := tan-1(x)
input s ← √2; h ← 1; M ← 30
for k = 0 to M do
    dk ← [f(s + h) - f(s - h)]/(2h)
    h ← h/2
end do
for k = 1 to M do
    rk ← dk + (dk - dk-1)/3
end do
output [k, dk, rk : 0 ≤ k ≤ M]
```

A few lines of output are shown here.

k	d _k	r _k
2	0.33719 385	0.33333 480
4	0.33357 477	0.33333 364
8	0.33332 825	0.33332 571
16	0.33203 125	0.33138 022
26	0.00000 000	0.00000 000

A comparison of this output with that of Example 2 shows that two more digits of accuracy have been obtained.

The example just given is artificial because we already know the correct result, and we are using this knowledge to determine which value in the third column is most accurate. In a realistic situation, we would count the number of significant figures lost in computing d_k . Obviously, r_k cannot have more significant figures than d_k . The moral here is *not* to assume that r_k becomes more accurate as k increases indefinitely. ■

It has probably occurred to the reader that what was done to Equation (15) can now be done (with suitable modifications) to Equation (17). The work involved is as follows: Put

$$\psi(h) = \frac{4}{3}\varphi(h/2) - \frac{1}{3}\varphi(h)$$

in Equation (17). Then

$$\begin{aligned} L &= \psi(h) + b_4h^4 + b_6h^6 + \cdots \\ L &= \psi(h/2) + b_4h^4/16 + b_6h^6/64 + \cdots \end{aligned}$$

Now

$$\begin{array}{r} L = \psi(h) + b_4h^4 + b_6h^6 + \cdots \\ 16L = 16\psi(h/2) + b_4h^4 + b_6h^6/4 + \cdots \\ \hline 15L = 16\psi(h/2) - \psi(h) - 3b_6h^6/4 - \cdots \end{array}$$

Thus, we have

$$L = \frac{16}{15}\psi(h/2) - \frac{1}{15}\psi(h) - b_6h^6/20 - \dots \quad (18)$$

Again, we can repeat this process by putting

$$\theta(h) = \frac{16}{15}\psi(h/2) - \frac{1}{15}\psi(h)$$

in Equation (18) so that

$$L = \theta(h) + c_6h^6 + c_8h^8 + \dots$$

In a manner similar to that used above, it follows that

$$L = \frac{64}{63}\theta(h/2) - \frac{1}{63}\theta(h) - 3c_8h^8/252 - \dots$$

As a matter of fact, any number of steps can be carried out to obtain formulas of increasing precision. The complete algorithm, allowing for M steps of **Richardson extrapolation algorithm**, is given next:

1. Select a convenient h (say $h = 1$) and compute the $M + 1$ numbers

$$D(n, 0) = \varphi(h/2^n) \quad (0 \leq n \leq M)$$

2. Compute additional quantities by the formula

$$D(n, k) = \frac{4^k}{4^k - 1} D(n, k - 1) - \frac{1}{4^k - 1} D(n - 1, k - 1) \quad (19)$$

Here $k = 1, 2, \dots, M$ and $n = k, k + 1, \dots, M$.

Observe that $D(0, 0) = \varphi(h)$, $D(1, 0) = \varphi(h/2)$, and $D(1, 1) = \psi(h)$. The quantities $D(n, 1)$ conform to Equation (17) with h repeatedly replaced by $h/2$. Similarly, $D(n, 2)$ corresponds to Equation (18), and so on. It is clear from our hypotheses and calculations that

$$D(n, 0) = L + \mathcal{O}(h^2)$$

$$D(n, 1) = L + \mathcal{O}(h^4)$$

$$D(n, 2) = L + \mathcal{O}(h^6)$$

$$D(n, 3) = L + \mathcal{O}(h^8)$$

The general result, proved in the next theorem, is that

$$D(n, k - 1) = L + \mathcal{O}(h^{2k}) \quad \text{as } h \rightarrow 0$$

At this point, let us reemphasize that the analysis can be applied to *any* numerical process that obeys Equation (15). The equation just preceding Equation (15) gives only one example of how it may arise in a specific numerical process.

If Equation (15) is valid, then the Richardson algorithm is effective, in the sense that successive columns in the $D(n, k)$ -array will exhibit higher-order convergence. This is the content of the next theorem.

THEOREM 1 Theorem on Richardson Extrapolation

The quantities $D(n, k)$ defined in the Richardson extrapolation algorithm obey an equation of the form

$$D(n, k - 1) = L + \sum_{j=k}^{\infty} A_{jk} (h/2^n)^{2j} \quad (20)$$

Proof When $k = 1$, we verify Equation (20) by using the definition of $D(n, 0)$ and Equation (15):

$$D(n, 0) = \varphi(h/2^n) = L - \sum_{j=1}^{\infty} a_{2j} (h/2^n)^{2j}$$

Thus, we can let $A_{j1} = -a_{2j}$. Now proceed by induction on k . We assume that Equation (20) is valid for some $k - 1$, and on that basis we prove it for k . From Equations (19) and (20), we have

$$D(n, k) = \frac{4^k}{4^k - 1} \left[L + \sum_{j=k}^{\infty} A_{j,k} \left(\frac{h}{2^n} \right)^{2j} \right] - \frac{1}{4^k - 1} \left[L + \sum_{j=k}^{\infty} A_{j,k} \left(\frac{h}{2^{n-1}} \right)^{2j} \right]$$

This can be simplified to the following:

$$D(n, k) = L + \sum_{j=k}^{\infty} A_{j,k} \left[\frac{4^k - 4^j}{4^k - 1} \right] \left(\frac{h}{2^n} \right)^{2j} \quad (21)$$

Thus, $A_{j,k+1}$ should be defined by

$$A_{j,k+1} = A_{j,k} \left[\frac{4^k - 4^j}{4^k - 1} \right]$$

Notice that $A_{k,k+1} = 0$, and thus Equation (21) can be written as

$$D(n, k) = L + \sum_{j=k+1}^{\infty} A_{j,k+1} (h/2^n)^{2j} \quad \blacksquare$$

The formulas given for $D(n, 0)$ and $D(n, k)$ allow us to construct a triangular array:

$$\begin{array}{ccccccc} D(0, 0) & & & & & & \\ D(1, 0) & D(1, 1) & & & & & \\ D(2, 0) & D(2, 1) & D(2, 2) & & & & \\ \vdots & \vdots & \vdots & \ddots & & & \\ D(M, 0) & D(M, 1) & D(M, 2) & \cdots & D(M, M) & & \end{array}$$

An algorithm to generate this triangular array is given next. In it, we refer to a function φ that must be made available to the code separately (i.e., by a subprogram or procedure).

```
input h, M
for n = 0 to M do
    D(n, 0) ← φ(h/2^n)
end do
for k = 1 to M do
    for n = k to M do
        D(n, k) ← D(n, k - 1) + [D(n, k - 1) - D(n - 1, k - 1)]/(4^k - 1)
    end do
end do
output D(n, k)  (0 ≤ n ≤ M, 0 ≤ k ≤ n)
```

EXAMPLE 6 Recompute the derivative in Example 5 using the complete Richardson extrapolation algorithm, just described.

Solution Some output from this algorithm, using Example 2, is shown here:

<i>n</i>	<i>D</i> (<i>n</i> , 0)	<i>D</i> (<i>n</i> , 1)	<i>D</i> (<i>n</i> , 2)	<i>D</i> (<i>n</i> , 3)	<i>D</i> (<i>n</i> , 4)
0	0.39269 91				
1	0.34877 10	0.33412 83			
2	0.33719 38	0.33333 48	0.33328 19		
3	0.33429 81	0.33333 29	0.33333 28	0.33333 36	
4	0.33357 48	0.33333 36	0.33333 37	0.33333 37	0.33333 37

This algorithm obtains approximately the same precision as in Example 5. Eventually it, too, will produce meaningless results because of subtractive cancellation. ■

In recent years, software tools have been developed for **automatic differentiation**. Bischof, Carle, Khademi, and Mauer [1994] have written a software system called ADIFOR that evaluates derivatives by applying the chain rule to elementary instructions. One of their goals is to generate efficient derivative codes that require little human effort, consistent with a *black-box* piece of software. For an introduction to techniques for evaluating derivatives of functions via computer programs, see Griewank and Corliss [1991], for example.

PROBLEMS 7.1

- 1. Carry out in all detail the derivation of Equation (11).
- 2. Let *f* be a polynomial of degree *n*. If the values of *f* at *n* points are known, can *f*'(*c*) or $\int_a^b f(x) dx$ be estimated with confidence? Your answers will depend on *c*, *a*, and *b*.
- 3. Let a numerical process be described by

$$L = \varphi(h) + \sum_{j=1}^{\infty} a_j h^j$$

Explain how Richardson extrapolation will work in this case. Prove an analogue of Theorem 1 in this section, on Richardson extrapolation, for this case.

- 4. Give the explicit form of Equation (11) in these cases:

- a. $n = 0$: $\alpha = 0$
- b. $n = 1$: $\alpha = 0$ and $\alpha = 1$
- c. $n = 2$: $\alpha = 0$ and $\alpha = 2$

5. Formula (9) for $f''(x)$ is often used in the numerical solution of differential equations. By adding the Taylor series for $f(x+h)$ and for $f(x-h)$, show that the error in this formula has the form $\sum_{n=1}^{\infty} a_{2n} h^{2n}$. Determine the coefficients a_{2n} explicitly. Also derive the error term given in Equation (9).
6. Derive the following two formulas for approximating derivatives and show that they are both $\mathcal{O}(h^4)$ by establishing their error terms:

$$f'(x) \approx \frac{1}{12h} [-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)]$$

$$f''(x) \approx \frac{1}{12h^2} [-f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h)]$$

7. Derive the following two formulas for approximating the third derivative. Find their error terms. Which formula is more accurate?

$$f'''(x) \approx \frac{1}{h^3} [f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)]$$

$$f'''(x) \approx \frac{1}{2h^3} [f(x+2h) - 2f(x+h) + 2f(x-h) - f(x-2h)]$$

8. (Continuation) Carry out the instructions in the preceding problem for the following fourth-derivative formulas:

$$f^{(4)} \approx \frac{1}{h^4} [f(x+4h) - 4f(x+3h) + 6f(x+2h) - 4f(x+h) + f(x)]$$

$$f^{(4)} \approx \frac{1}{h^4} [f(x+2h) - 4f(x+h) + 6f(x) - 4f(x-h) + f(x-2h)]$$

9. Show that in Richardson extrapolation,

$$D(2, 2) = \frac{16}{15} \psi(h/2) - \frac{1}{15} \psi(h)$$

10. Show how to use Richardson extrapolation employing x_n and x_{n^2} if

$$L = x_n + a_1 n^{-1} + a_2 n^{-2} + a_3 n^{-3} + \dots$$

11. Prove or disprove:

- a. If $L - x_n = \mathcal{O}(n^{-1})$, then $L - (2x_{2n} - x_n) = \mathcal{O}(n^{-2})$.
- b. If $L - x_n = \mathcal{O}(n^{-1})$, then $L - x_{n^2} = \mathcal{O}(n^{-2})$.

Discuss the numerical consequences of this problem.

12. Show how to use Richardson extrapolation if

$$L = \varphi(h) + a_1 h + a_3 h^3 + a_5 h^5 + \dots$$

13. Suppose that $L = \lim_{h \rightarrow 0} f(h)$ and that $L - f(h) = c_6 h^6 + c_9 h^9 + \dots$. What combination of $f(h)$ and $f(h/2)$ should be the best estimate of L ?
14. Using Taylor series, derive the error term for the approximation

$$f'(x) \approx \frac{1}{2h} [-3f(x) + 4f(x+h) - f(x+2h)]$$

15. Derive a numerical differentiation formula of order $\mathcal{O}(h^4)$ by applying Richardson extrapolation to

$$f'(x) = \frac{1}{2h} [f(x+h) - f(x-h)] - \frac{h^2}{6} f'''(x) - \frac{h^4}{120} f^{(5)}(x) - \dots$$

Give the error term of order $\mathcal{O}(h^4)$.

16. Using Taylor series expansions, derive the error term for the formula

$$f''(x) \approx \frac{1}{h^2} [f(x) - 2f(x+h) + f(x+2h)]$$

17. Establish a formula of the form

$$f''_n \approx \frac{1}{h^2} [Af_{n+3} + Bf_{n+2} + Cf_{n+1} + Df_n]$$

Here $f_{n+i} = f(x_n + ih)$.

18. Derive the approximation

$$f'(x_n) \approx \frac{3f(x_n) - 4f(x_{n-1}) + f(x_{n-2}))}{3x_n - 4x_{n-1} + x_{n-2}}$$

and show that the error term is $\mathcal{O}(h^2)$ as $h \rightarrow 0$. Here $f_{n+i} = f(x_n + ih)$.

COMPUTER
PROBLEMS 7.1

1. Program the algorithm in the text in which Richardson extrapolation is used repeatedly to estimate $f'(x)$. Test your program on the following:
 - a. $\ln x$ at $x = 3$
 - b. $\tan x$ at $x = \sin^{-1}(0.8)$
 - c. $\sin(x^2 + \frac{1}{3}x)$ at $x = 0$
2. Write and test an algorithm for computing $f''(x)$ using Formula (9) together with repeated Richardson extrapolation.
3. In a typical use of Formula (1), the roundoff error (due mainly to subtractive cancellation) will behave like αh^{-1} . Test this assertion (and estimate α) on some examples. A computer program using multiple precision may be needed to simulate the computations.
4. Derive and test an algorithm for approximating f' and f'' using cubic splines.
5. Using the approximation in Problem 7.1.18 above, develop an iterative formula analogous to the secant method. Test the numerical behavior of this method on several functions, and compare it with the secant method.

7.2 Numerical Integration Based on Interpolation

Numerical integration is the process of producing a numerical value for the integration of a function over a set. Here are some examples of integrals that might be

calculated by suitable computer routines:

$$\begin{aligned} &\int_0^2 e^{-x^2} dx \\ &\int_0^1 \int_0^1 \sin(xye^x) dx dy \\ &\int_0^1 \int_{x^2}^x \tan(xy^2) dy dx \\ &\int_0^\pi \cos(3 \cos \theta) d\theta \end{aligned}$$

These integration problems are not amenable to the techniques learned in elementary calculus. Those techniques depend on **antidifferentiation**. Thus, to find the value of

$$\int_a^b f(x) dx$$

by calculus, we first produce a function F with the property that $F' = f$. Then we have

$$\int_a^b f(x) dx = F(b) - F(a)$$

For example, we have

$$\int_1^4 x^2 dx = \left. \frac{1}{3} x^3 \right|_1^4 = \frac{1}{3} 4^3 - \frac{1}{3} = 21$$

The function F given by $F(x) = (1/3)x^3$ is an antiderivative of the function f given by $f(x) = x^2$. Many elementary functions do not have simple antiderivatives. A good example is $f(x) = e^{x^2}$. An antiderivative of this function is

$$F(x) = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)k!}$$

(See Equation (3) in Section 6.7, p. 390, for the source of this equation.)

One powerful stratagem for numerically computing the integral

$$\int_a^b f(x) dx \tag{1}$$

is to replace f by another function g that approximates f well and is easily integrated. Then we simply say to ourselves that from $f \approx g$, it follows that

$$\int_a^b f(x) dx \approx \int_a^b g(x) dx$$

It should now occur to the reader that polynomials are good candidates for the function g , and indeed g can be a polynomial that *interpolates* to f at a certain set of nodes. Of course, a polynomial approximation to f can be obtained in other ways, for example, by truncating a Taylor series. Again, the reader should recall what was similarly done in Section 6.7 (p. 390). Thus, for example, we have

$$\int_0^1 e^{x^2} dx \approx \int_0^1 \sum_{k=0}^n \frac{x^{2k}}{k!} dx = \sum_{k=0}^n \frac{1}{(2k+1)k!}$$

However, it is desirable to have general procedures that require only *evaluations* of the integrand. Methods based on interpolation fulfill this desideratum. Spline functions also can be used to interpolate f , and integrals of spline functions are easily computed.

Integration via Polynomial Interpolation

We take up polynomial interpolation now. Suppose that we want to evaluate the integral in (1). We can select nodes $x_0, x_1, \dots, x_n$ in $[a, b]$ and set up a Lagrange interpolation process as in Section 6.1 (p. 312). Define

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \quad (0 \leq i \leq n)$$

These are the **fundamental polynomials for interpolation**. The polynomial of degree at most n that interpolates f at the nodes is

$$p(x) = \sum_{i=0}^n f(x_i) \ell_i(x) \tag{2}$$

Then, as mentioned previously, we write simply

$$\int_a^b f(x) dx \approx \int_a^b p(x) dx = \sum_{i=0}^n f(x_i) \int_a^b \ell_i(x) dx$$

In this way, we obtain a formula that can be used on *any* f . It reads as follows:

$$\int_a^b f(x) dx \approx \sum_{i=0}^n A_i f(x_i) \tag{3}$$

where

$$A_i = \int_a^b \ell_i(x) dx$$

A formula of the form in (3) is called a **Newton-Cotes formula** if the nodes are equally spaced.

Trapezoid Rule

The simplest case results if $n = 1$ and the nodes are $x_0 = a$ and $x_1 = b$. Then the fundamental polynomials for interpolation are

$$\ell_0(x) = \frac{b-x}{b-a} \quad \ell_1(x) = \frac{x-a}{b-a}$$

Consequently,

$$A_0 = \int_a^b \ell_0(x) dx = \frac{1}{2}(b-a) = \int_a^b \ell_1(x) dx = A_1$$

The corresponding quadrature formula is

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

This is known as the **trapezoid rule**. It is exact for all $f \in \Pi_1$ (i.e., polynomials of degree at most 1). Moreover, its error term is

$$-\frac{1}{12}(b-a)^3 f''(\xi)$$

where $\xi \in (a, b)$. This can be determined by integrating the error term in the polynomial approximation $f(x) - p_1(x) = f''(\xi_x)(x-a)(x-b)/2$ and using the Mean-Value Theorem for Integrals. We shall see the trapezoid rule again, in Section 7.4 (p. 502), as a component of the Romberg integration method.

If the interval $[a, b]$ is partitioned like this:

$$a = x_0 < x_1 < \cdots < x_n = b$$

then the trapezoid rule can be applied to each subinterval. Here the nodes are not necessarily uniformly spaced. Thus, we obtain the **composite trapezoid rule**:

$$\begin{aligned} \int_a^b f(x) dx &= \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x) dx \\ &\approx \frac{1}{2} \sum_{i=1}^n (x_i - x_{i-1}) [f(x_{i-1}) + f(x_i)] \end{aligned}$$

(A **composite rule** is one obtained by applying an integration formula for a single interval to each subinterval of a partitioned interval.) The composite trapezoid rule also arises if the integrand f is replaced by an interpolating spline function of degree 1—that is, a *broken line*.

With uniform spacing $h = (b-a)/n$ and $x_i = a + ih$, the **composite trapezoid rule** takes the form

$$\int_a^b f(x) dx \approx \frac{h}{2} \left[f(a) + 2 \sum_{i=1}^{n-1} f(a + ih) + f(b) \right]$$

or

$$\int_a^b f(x) dx \approx h \sum_{i=0}^{n''} f(a + ih) \tag{4}$$

where the double prime on the summation indicates that the first and last terms in the sum are to be halved. The error term for the composite trapezoid rule is

$$-\frac{1}{12}(b-a)h^2 f''(\xi)$$

where $\xi \in (a, b)$. It is obtained from summing the error term on each subinterval and using the fact that a point ξ exists in $[a, b]$ for which $f''(\xi) = (1/n) \sum_{i=1}^n f''(\xi_i)$, where $\xi_i \in (x_{i-1}, x_i)$ and $(1/n) = (b-a)/h$ —that is, the average value.

EXAMPLE 1 If we take $n = 2$ and $[a, b] = [0, 1]$ in the Newton-Cotes procedure, we obtain another formula:

$$\int_0^1 f(x) dx \approx \frac{1}{6}f(0) + \frac{2}{3}f\left(\frac{1}{2}\right) + \frac{1}{6}f(1) \tag{5}$$

Derive this rule by using Formula (3).

Solution The three fundamental polynomials for nodes 0, $\frac{1}{2}$, and 1 are

$$\ell_0(x) = 2\left(x - \frac{1}{2}\right)(x - 1) \qquad \ell_1(x) = -4x(x - 1) \qquad \ell_2(x) = 2x\left(x - \frac{1}{2}\right)$$

Then

$$A_0 = \int_0^1 \ell_0(x) dx = \frac{1}{6}$$

and so forth. ■

Method of Undetermined Coefficients

From the manner in which Formula (3) was derived, we see at once that it is *exact* for all polynomials of degree $\leq n$. On the other hand, suppose that Formula (3) is given to us, and we are told only that it is exact for all polynomials of degree $\leq n$. Is the following equation true?

$$A_i = \int_a^b \ell_i(x) dx$$

The answer is yes, and we see this from the fact that Formula (3) should correctly integrate any ℓ_j . Hence,

$$\int_a^b \ell_j(x) dx = \sum_{i=0}^n A_i \ell_j(x_i) = A_j$$

Here, of course, we used two essential properties of the fundamental polynomials: ℓ_j is a polynomial of degree at most n and $\ell_i(x_j) = \delta_{ij}$.

The remarks just made enable us to arrive at formulas like (3) by the **method of undetermined coefficients**. To illustrate, let us derive Equation (5) in this way. We seek a formula

$$\int_0^1 f(x) dx \approx A_0 f(0) + A_1 f\left(\frac{1}{2}\right) + A_2 f(1)$$

that will be exact for all polynomials of degree ≤ 2 . By using as trial functions the polynomials $f(x) = 1, x$, and x^2 , in order, we get

$$\begin{aligned} 1 &= \int_0^1 dx = A_0 + A_1 + A_2 \\ \frac{1}{2} &= \int_0^1 x dx = \frac{1}{2}A_1 + A_2 \\ \frac{1}{3} &= \int_0^1 x^2 dx = \frac{1}{4}A_1 + A_2 \end{aligned}$$

The solution of this system of three simultaneous equations is $A_0 = 1/6$, $A_1 = 2/3$, and $A_2 = 1/6$. Since the formula is linear, it will produce exact values of integrals for any quadratic polynomial, $f(x) = c_0 + c_1x + c_2x^2$.

Simpson's Rule

Similar calculations for an arbitrary interval $[a, b]$ lead to the familiar **Simpson's rule**:

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] \quad (6)$$

From the manner in which it was derived, we know that Simpson's rule is exact for all polynomials of degree ≤ 2 . It is, unexpectedly, exact for all polynomials of degree ≤ 3 . (See Problem 7.1.2, p. 476.)

The error term associated with Simpson's rule is

$$-\frac{1}{90}[(b-a)/2]^5 f^{(4)}(\xi)$$

for some $\xi \in (a, b)$. The derivation of this error term is most easily carried out using the Peano Kernel Theorem, as suggested in Problem 7.6.5 (p. 519). We can see that the error is $\mathcal{O}(h^5)$, by proceeding as follows. If $h = (b-a)/2$, this numerical integration formula takes the form

$$\int_a^{a+2h} f(x) dx \approx \frac{h}{3} [f(a) + 4f(a+h) + f(a+2h)]$$

Using Taylor's Theorem from Section 1.1 (p. 6), we can write the right-hand side as

$$2hf(a) + 2h^2 f'(a) + \frac{4}{3}h^3 f''(a) + \frac{2}{3}h^4 f'''(a) + \frac{100}{3 \cdot 5!}h^5 f^{(4)}(a) + \dots$$

Now let

$$F(x) = \int_a^x f(t) dt$$

By the Fundamental Theorem of Calculus, $F' = f$, and with Taylor's Theorem we can write the left-hand side in Simpson's rule as $F(a + 2h)$ or

$$2hf(a) + 2h^2 f'(a) + \frac{4}{3}h^3 f''(a) + 23h^4 f'''(a) + \frac{32}{5!}h^5 f^{(4)}(a) + \dots$$

Combining these two expansions, we have

$$\int_a^{a+2h} f(x) dx = \frac{h}{3}[f(a) + 4f(a+h) + f(a+2h)] - \frac{1}{90}h^5 f^{(4)}(a) - \dots$$

A **composite Simpson's rule** using an even number of subintervals is often used. Let n be even, and set

$$x_i = a + ih \quad h = (b - a)/n \quad (0 \leq i \leq n)$$

Then

$$\begin{aligned} \int_a^b f(x) dx &= \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \dots + \int_{x_{n-2}}^{x_n} f(x) dx \\ &= \sum_{i=1}^{n/2} \int_{x_{2i-2}}^{x_{2i}} f(x) dx \end{aligned}$$

Now let Simpson's rule (6) be applied to each subinterval, resulting in the formula

$$\int_a^b f(x) dx \approx \frac{h}{3} \sum_{i=1}^{n/2} [f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})]$$

The right-hand side of this formula should be computed as follows, to avoid repetition of terms:

$$\frac{h}{3} \left[f(x_0) + 2 \sum_{i=2}^{n/2} f(x_{2i-2}) + 4 \sum_{i=1}^{n/2} f(x_{2i-1}) + f(x_n) \right]$$

The error term for this formula is

$$-\frac{1}{180}(b-a)h^4 f^{(4)}(\xi)$$

for some $\xi \in (a, b)$.

General Integration Formulas

The procedure that led to the Newton-Cotes formulas can be used to produce more general integration formulas of the type

$$\int_a^b f(x)w(x) dx \approx \sum_{i=0}^n A_i f(x_i)$$

where w can be any fixed **weight function**. The only modification necessary is to replace Equation (3) by

$$A_i = \int_a^b \ell_i(x) w(x) dx$$

This is illustrated in the next example.

EXAMPLE 2 Find a formula

$$\int_{-\pi}^{\pi} f(x) \cos x dx \approx A_0 f\left(-\frac{3}{4}\pi\right) + A_1 f\left(-\frac{1}{4}\pi\right) + A_2 f\left(\frac{1}{4}\pi\right) + A_3 f\left(\frac{3}{4}\pi\right)$$

that is exact when f is a polynomial of degree 3.

Solution Clearly, a polynomial of degree 3 is a linear combination of the four monomials 1, x , x^2 , and x^3 . Thus, we can determine the coefficients by substituting $f(x) = x^j$ ($0 \leq j \leq 3$) and solving the resulting four linear equations. By symmetry, $A_0 = A_3$ and $A_1 = A_2$; thus, the work can be simplified:

$$0 = \int_{-\pi}^{\pi} 1 \cos x dx = 2A_0 + 2A_1$$

$$-4\pi = \int_{-\pi}^{\pi} x^2 \cos x dx = 2A_0 \left(\frac{3}{4}\pi\right)^2 + 2A_1 \left(\frac{1}{4}\pi\right)^2$$

The solution is $A_1 = A_2 = -A_0 = -A_3 = 4/\pi$, and the formula is

$$\int_{-\pi}^{\pi} f(x) \cos x dx \approx \frac{4}{\pi} \left[-f\left(-\frac{3}{4}\pi\right) + f\left(-\frac{1}{4}\pi\right) + f\left(\frac{1}{4}\pi\right) - f\left(\frac{3}{4}\pi\right) \right] \quad \blacksquare$$

Change of Intervals

From a formula for numerical integration on one interval, we can derive a formula for any other interval by making a linear change of variable. If the first formula is exact for polynomials of a certain degree, the same will be true of the second. Let us see how this is accomplished.

Suppose that a numerical integration formula is given:

$$\int_c^d f(t) dt \approx \sum_{i=0}^n A_i f(t_i) \quad (7)$$

We do not care where this formula comes from; however, let us assume that it is exact for all polynomials of degree $\leq m$. If a formula is needed for some other interval, say $[a, b]$, we first define a linear function λ of t such that if t traverses $[c, d]$, $\lambda(t)$ will traverse $[a, b]$. The function λ is given explicitly by

$$\lambda(t) = \frac{b-a}{d-c}t + \frac{ad-bc}{d-c} \quad (8)$$

Now in the integral

$$\int_a^b f(x) dx$$

we make the change of variable $x = \lambda(t)$. Then $dx = \lambda'(t) dt = (b-a)(d-c)^{-1} dt$, and so we have

$$\begin{aligned} \int_a^b f(x) dx &= \frac{b-a}{d-c} \int_c^d f(\lambda(t)) dt \\ &\approx \frac{b-a}{d-c} \sum_{i=0}^n A_i f(\lambda(t_i)) \end{aligned}$$

Hence, we have

$$\int_a^b f(x) dx \approx \frac{b-a}{d-c} \sum_{i=0}^n A_i f\left(\frac{b-a}{d-c} t_i + \frac{ad-bc}{d-c}\right) \quad (9)$$

Observe that, because λ is linear, $f(\lambda(t))$ is a polynomial in t if f is a polynomial, and the degrees are the same. Hence, the new formula is exact for polynomials of degree m . For Simpson's rule, this procedure yields Equation (6) as a consequence of Equation (5) using $\lambda(t) = (b-a)t + a$.

Error Analysis

To assess the error involved in the numerical integration Formula (3), we use the error term in polynomial interpolation. Recall, from Equation (13) in Section 6.1 (p. 315), that if p is the polynomial of degree $\leq n$ that interpolates f at $x_0, x_1, \dots, x_n$, and if $f^{(n+1)}$ is continuous, then

$$f(x) - p(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi_x) \prod_{i=0}^n (x - x_i) \quad (10)$$

Consequently,

$$\int_a^b f(x) dx - \sum_{i=0}^n A_i f(x_i) = \frac{1}{(n+1)!} \int_a^b f^{(n+1)}(\xi_x) \prod_{i=0}^n (x - x_i) dx \quad (11)$$

If $|f^{(n+1)}(x)| \leq M$ on $[a, b]$, then

$$\left| \int_a^b f(x) dx - \sum_{i=0}^n A_i f(x_i) \right| \leq \frac{M}{(n+1)!} \int_a^b \prod_{i=0}^n |x - x_i| dx \quad (12)$$

The choice of nodes that makes the right-hand side of this inequality as small as possible is known to be

$$x_i = \frac{a+b}{2} + \frac{b-a}{2} \cos \left[\frac{(i+1)\pi}{n+2} \right] \quad (0 \leq i \leq n)$$

If the interval is $[-1, 1]$, these nodes have the simpler form

$$x_i = \cos \left[\frac{(i+1)\pi}{n+2} \right] \quad (0 \leq i \leq n)$$

These are the zeros of the function

$$U_{n+1}(x) = \frac{\sin[(n+2)\theta]}{\sin \theta} \quad (x = \cos \theta) \quad (13)$$

The function U_{n+1} is known as a **Chebyshev polynomial of the second kind**. As defined in Equation (13), $U_{n+1}(x)$ is not a monic polynomial; that is, its leading coefficient is not 1. In fact, the coefficient of x^{n+1} in U_{n+1} is 2^{n+1} . Thus,

$$2^{-(n+1)}U_{n+1} = (x - x_0)(x - x_1) \cdots (x - x_n) \quad (14)$$

with $x_i = \cos[(i+1)\pi/(n+2)]$. A computation then shows that

$$\int_{-1}^1 |(x - x_0)(x - x_1) \cdots (x - x_n)| dx = 2^{-n} \quad (15)$$

Thus, with these nodes and the proper coefficients A_i , we have

$$\left| \int_{-1}^1 f(x) dx - \sum_{i=0}^n A_i f(x_i) \right| \leq \frac{M}{(n+1)! 2^n} \quad (16)$$

THEOREM 1 Theorem on Extremal Property, Chebyshev Polynomials of the Second Kind

Among the monic polynomials p of degree n , the one for which $\int_{-1}^1 |p(x)| dx$ is least is $2^{-n}U_n$.

Proof First, we prove the orthogonality relation:

$$\int_{-1}^1 U_m(x) \operatorname{sign}[U_n(x)] dx = 0 \quad (0 \leq m < n)$$

Let I denote this integral. Make the change of variable $\cos \theta = x$ to get

$$\begin{aligned} I &= \int_0^\pi \frac{\sin(m+1)\theta}{\sin \theta} \operatorname{sign} \left[\frac{\sin(n+1)\theta}{\sin \theta} \right] \sin \theta d\theta \\ &= \sum_{k=0}^n (-1)^k \int_{k\varphi}^{(k+1)\varphi} \sin(m+1)\theta d\theta \quad \varphi = \frac{\pi}{(n+1)} \\ &= (m+1)^{-1} \sum_{k=0}^n (-1)^{k+1} [\cos(m+1)(k+1)\varphi - \cos(m+1)k\varphi] \end{aligned}$$

Let $\alpha = (m + 1)\varphi + \pi$. Then, using $\operatorname{Re}(x)$ to signify the real part of x , we have

$$\begin{aligned}(m + 1)I &= \sum_{k=0}^n [\cos(k + 1)\alpha + \cos k\alpha] \\&= \operatorname{Re} \left\{ \sum_{k=0}^n [e^{i\alpha(k+1)} + e^{i\alpha k}] \right\} \\&= \operatorname{Re} \left\{ \frac{e^{i\alpha(n+2)} - e^{i\alpha} + e^{i\alpha n} - 1}{e^{i\alpha} - 1} \right\} \\&= \frac{\operatorname{Re}[(e^{-i\alpha} - 1)(e^{i\alpha(n+2)} - e^{i\alpha} + e^{i\alpha n} - 1)]}{|e^{i\alpha} - 1|^2}\end{aligned}$$

Here the numerator is

$$\begin{aligned}\operatorname{Re}(e^{i\alpha n} - e^{i\alpha(n+2)}) &= \cos n\alpha - \cos(n + 2)\alpha \\&= \cos[(n + 1)\alpha - \alpha] - \cos[(n + 1)\alpha + \alpha] \\&= \cos(k\pi - \alpha) - \cos(k\pi + \alpha) = 0\end{aligned}$$

where $k = m + n + 2$. To complete the proof, let p be a monic polynomial of degree n . It can be expressed as

$$p = 2^{-n}U_n + a_{n-1}U_{n-1} + \cdots + a_0U_0$$

Hence, by the orthogonality relation,

$$\begin{aligned}\int_{-1}^1 |p| \, dx &\geq \int_{-1}^1 p \operatorname{sign}[U_n] \, dx = 2^{-n} \int_{-1}^1 U_n \operatorname{sign}[U_n] \, dx \\&= 2^{-n} \int_{-1}^1 |U_n| \, dx\end{aligned}$$

■

The subject of numerical integration can be studied further in Davis and Rabinowitz [1984], Krylov [1962], and Ghizetti and Ossicini [1970].

PROBLEMS 7.2

1. Derive the Newton-Cotes formula for $\int_0^1 f(x) \, dx$ based on the nodes $0, \frac{1}{3}, \frac{2}{3}$, and 1 .
2. Prove (without using its error term) that Simpson's rule, Equation (6), correctly integrates all cubic polynomials.
3. Obtain Formula (6) from Formula (5) by a suitable change of variable.
4. Verify that the following formula is exact for polynomials of degree ≤ 4 :

$$\int_0^1 f(x) \, dx \approx \frac{1}{90} \left[7f(0) + 32f\left(\frac{1}{4}\right) + 12f\left(\frac{1}{2}\right) + 32f\left(\frac{3}{4}\right) + 7f(1) \right]$$

5. (Continuation) From the formula in the preceding problem, obtain a formula for $\int_a^b f(x) \, dx$ that is exact for all polynomials of degree 4.

6. (Continuation) Calculate $\ln 2$ approximately by applying the formula in the preceding problem to

$$\int_0^1 \frac{dt}{t+1}$$

Compare your answer to the correct value and compute the error.

7. Calculate $\int_0^1 e^{x^2} dx$ to eight-decimal-place accuracy by use of the series in the text.
8. Find the formula

$$\int_0^1 f(x) dx \approx A_0 f(0) + A_1 f(1)$$

that is exact for all functions of the form $f(x) = ae^x + b \cos(\pi x/2)$.

9. Find a formula of the form

$$\int_0^{2\pi} f(x) dx = A_1 f(0) + A_2 f(\pi)$$

that is exact for any function having the form

$$f(x) = a + b \cos x$$

Prove that the resulting formula is exact for any function of the form

$$f(x) = \sum_{k=0}^n [a_k \cos(2k+1)x + b_k \sin kx]$$

10. Use the Lagrange interpolation polynomial to derive the formula of the form

$$\int_0^1 f(x) dx \approx Af\left(\frac{1}{3}\right) + Bf\left(\frac{2}{3}\right)$$

Transform this formula to one for integration over $[a, b]$.

11. Using the polynomial of lowest order that interpolates $f(x)$ at x_1 and x_2 , derive a numerical integration formula for

$$\int_{x_0}^{x_3} f(x) dx$$

Do not assume uniform spacing. Here $x_0 < x_1 < x_2 < x_3$.

12. Derive a formula for approximating

$$\int_1^3 f(x) dx$$

in terms of $f(0)$, $f(2)$, and $f(4)$. It should be exact for all f in Π_2 .

13. Determine values for A , B , and C that make the formula

$$\int_0^2 xf(x) dx \approx Af(0) + Bf(1) + Cf(2)$$

exact for all polynomials of degree as high as possible. What is the maximum degree?

14. Derive the Newton-Cotes formula for

$$\int_0^1 f(x) dx$$

based on the Lagrange interpolation polynomial at the nodes -2 , -1 , and 0 . Apply this result to evaluate the integral when $f(x) = \sin \pi x$.

15. Calculate

$$\int_0^{10^{-2}} \left(\frac{\sin x}{x} \right) dx$$

to seven-decimal-place accuracy using a series.

16. We intend to use $\int_0^1 p(x) dx$ as an estimate of $\int_0^1 f(x) dx$, where p is a polynomial of degree n that interpolates f at nodes $x_0, x_1, \dots, x_n$ in $[0, 1]$. Assume that $|f^{(n+1)}(x)| < M$ on $[0, 1]$. What upper bound can be given for the error $|\int_0^1 f(x) dx - \int_0^1 p(x) dx|$ if nothing is known about the location of the nodes? Can you find the best upper bound?
17. Determine the composite numerical integration rule based on the simple **right-side rect-angle rule**:

$$\int_0^1 f(x) dx \approx f(1)$$

Assume unequal spacing $a = x_0 < x_1 < \dots < x_n = b$.

18. Derive the composite rule for $\int_a^b f(x) dx$ based on the midpoint rule

$$\int_{-1}^1 f(x) dx \approx 2f(0)$$

Give formulas for unequal spacing and equal spacing of nodes.

19. (Continuation) The **midpoint rule** over the interval $[x_{i+1}, x_{i-1}]$ is given by

$$\int_{x_{i-1}}^{x_{i+1}} f(x) dx = (x_{i+1} - x_{i-1}) f(x_i)$$

Determine the composite midpoint rule over the interval $[a, b]$ with uniform spacing of $h = (b - a)/n$ such that $x_i = a + ih$ for $i = 0, 1, 2, \dots, n$ (n is even).

20. Determine the integration rule for $\int_a^b f(x) dx$ based on the Gaussian quadrature rule

$$\int_{-1}^1 f(x) dx \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

21. There are two Newton-Cotes formulas for $n = 2$ and $[a, b] = [0, 1]$; namely,

$$\int_0^1 f(x) dx \approx af(0) + bf\left(\frac{1}{2}\right) + cf(1)$$

$$\int_0^1 f(x) dx \approx \alpha f\left(\frac{1}{4}\right) + \beta f\left(\frac{1}{2}\right) + \gamma f\left(\frac{3}{4}\right)$$

Which is better?

22. Is there a formula of the form

$$\int_0^1 f(x) dx \approx \alpha[f(x_0) + f(x_1)]$$

that correctly integrates all quadratic polynomials?

23. Prove that if the formula

$$\int_{-1}^1 f(x) dx \approx \sum_{i=0}^n A_i f(x_i) \quad (n \text{ is even})$$

is exact for all polynomials of degree n , and if the nodes are symmetrically placed about the origin, then the formula is exact for all polynomials of degree $n + 1$.

24. Let n be even. Show how the composite Simpson rule with $2n$ equally spaced nodes can be computed from the case of n equally spaced nodes with a minimum amount of additional work.
25. In Example 2, symmetry was used to simplify the calculations. Give a proof that $A_0 = A_3$ and $A_1 = A_2$.
26. Prove that the **Chebyshev polynomials of the second kind** are generated recursively by

$$\begin{cases} U_0(x) = 1 & U_1(x) = 2x \\ U_{n+1} = 2xU_n - U_{n-1} & (n \geq 1) \end{cases}$$

27. (Continuation) Prove the orthogonality relation

$$\int_{-1}^1 U_n(x)U_m(x)\sqrt{1-x^2} dx = \delta_{nm} \frac{\pi}{2}$$

28. (Continuation) Prove this relation between Chebyshev polynomials of the first and second kinds: $T'_n = nU_{n-1}$.
29. Establish error terms for the trapezoid rule and the composite trapezoid rule.
30. Fill in the details in establishing the error term for Simpson's rule and the composite Simpson's rule.
31. Determine the minimum number of subintervals needed to approximate

$$\int_1^2 (x + e^{-x^2}) dx$$

to an accuracy of at least $\frac{1}{2} \times 10^{-7}$, using the trapezoid rule.

COMPUTER PROBLEMS 7.2

1. Write a computer program that computes $\int_0^x e^{-t^2} dt$ by summing an appropriate Taylor series until the individual terms fall below 10^{-8} in magnitude. Test your program by calculating the values of this integral for $x = 0.0, 0.1, 0.2, \dots, 1.0$.
2. Write a computer program that estimates $\int_a^b f(x) dx$ by $\int_a^b S(x) dx$, where S is the natural cubic spline having knots $a + ih$ and interpolating f at these knots. Here $0 \leq i \leq n$ and $h = (b - a)/n$. First obtain a formula for

$$\int_{t_0}^{t_n} S(x) dx$$

starting with Equation (7) in Section 6.4 (p. 351). Then write the subprogram to compute this. Test your code on these well-known integrals:

- a. $\frac{4}{\pi} \int_0^1 (1 + x^2)^{-1} dx$

- b. $\frac{1}{\ln 3} \int_1^3 x^{-1} dx$

Use $n = 4, 8$, and 16 in both examples. (See Computer Problem 6.4.3, p. 364.)

3. Using a symbolic manipulation program, carry out the following exercises:
 - a. Find the indefinite integral $\int (\cos x)^{-14} dx$.

- b. Find the definite integral $\int_0^1 \log(\log x) dx$.
 c. Obtain the numerical value of $\int_0^1 \sqrt{1 + \sin^3 x} dx$.

7.3 Gaussian Quadrature

In the preceding section, we saw how to create quadrature formulas of the type

$$\int_a^b f(x) dx \approx \sum_{i=0}^n A_i f(x_i) \quad (1)$$

that are *exact* for polynomials of degree $\leq n$. In these formulas, the choice of nodes $x_0, x_1, \dots, x_n$ was made *a priori*. Once the nodes were fixed, the coefficients were determined uniquely from the requirement that Formula (1) must be an equality for all $f \in \Pi_n$.

It is natural to ask whether some choices of nodes are better than others in Formula (1). For example, a particular set of nodes may exist for which the resulting coefficients A_i are all equal to each other. If $A_i = c$ for $0 \leq i \leq n$, less arithmetic is involved in using Equation (1), because it now has this simpler form:

$$\int_a^b f(x) dx \approx c \sum_{i=0}^n f(x_i) \quad (2)$$

(This effects a reduction from $n + 1$ to 1 in the number of multiplications.) Formulas of type (2) exist only for $n = 0, 1, 2, 3, 4, 5, 6$, and 8. They are known as **Chebyshev's quadrature formulas**. The one corresponding to $n = 4$ is

$$\int_{-1}^1 f(x) dx \approx \frac{2}{5} [f(-\alpha) + f(-\beta) + f(0) + f(\beta) + f(\alpha)] \quad (3)$$

where

$$\alpha = \sqrt{(5 + \sqrt{11})/12} \approx 0.83249\ 74870\ 00982$$

$$\beta = \sqrt{(5 - \sqrt{11})/12} \approx 0.37454\ 14095\ 53581$$

The method of undetermined coefficients can be used to obtain the nodes α and β . Also, it can be shown that the formula is exact for polynomials of degree ≤ 5 . Somewhat more complicated formulas with equal coefficients can also be given. As an example, we cite

$$\int_{-1}^1 f(x) dx \approx \frac{\pi}{n} \sum_{i=1}^n F\left(\cos \frac{2i-1}{2n} \pi\right) \quad (4)$$

in which $F(x) = f(x)\sqrt{1-x^2}$. This is known as **Hermite's quadrature formula**. It is exact for a certain linear space of dimension $2n$, namely,

$$G = \{pw : p \in \Pi_{2n-1}\} \quad w(x) = (1-x^2)^{-1/2}$$

It is not likely that we will save computing effort when this formula is used. (Why?) On the other hand, with only n evaluations of f , we have a formula that is exact on a linear space of dimension $2n$. Pursuing this goal systematically leads us to the *Gaussian* quadrature formulas.

Gaussian Quadrature

The theory can be formulated for quadrature rules of a slightly more general form; namely,

$$\int_a^b f(x)w(x) dx \approx \sum_{i=0}^n A_i f(x_i) \quad (5)$$

where w is a fixed positive **weight function**. The case when $w(x) \equiv 1$ is, naturally, of special importance. Let us assume that Formula (5) is exact for $f \in \Pi_n$. As we learned in the preceding section, this is the case if and only if

$$A_i = \int_a^b w(x) \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} dx \quad (6)$$

Because there are $n + 1$ coefficients A_i and $n + 1$ nodes x_i at our disposal with no *a priori* restrictions on them, we suspect that quadrature formulas of the form in Equation (5) that will be exact for polynomials of degree $\leq 2n + 1$ can be discovered. We now show that this is indeed true.

The idea, which originated with Carl Friedrich Gauss (1777–1855), is to use the variability of the nodes to force the quadrature Formulas (5) and (6) to be exact for all polynomials of degree $2n + 1$. Here is the theorem that discloses where the nodes should be placed.

THEOREM 1 Theorem on Gaussian Quadrature

Let w be a positive weight function and let q be a nonzero polynomial of degree $n + 1$ that is *w-orthogonal* to Π_n in the sense that for any $p \in \Pi_n$, we have

$$\int_a^b q(x)p(x)w(x) dx = 0 \quad (7)$$

If $x_0, x_1, \dots, x_n$ are the zeros of q , then the quadrature Formula (5) with coefficients given by Equation (6) will be exact for all $f \in \Pi_{2n+1}$.

Proof Let $f \in \Pi_{2n+1}$. Divide f by q , obtaining a quotient p and remainder r . Then

$$f = qp + r \quad (p, r \in \Pi_n)$$

Consequently, $f(x_i) = r(x_i)$. Using Equation (7) and the fact that Equation (5) is exact for elements of Π_n , we have

$$\int_a^b f w dx = \int_a^b r w dx = \sum_{i=0}^n A_i r(x_i) = \sum_{i=0}^n A_i f(x_i) \quad \blacksquare$$

It turns out that the roots of q are **simple roots** and lie in the interior of the interval $[a, b]$. (In particular, they are real, not imaginary.) This follows immediately from the following theorem.

THEOREM 2 Theorem on Number of Sign Changes

Let w be a positive weight function in $C[a, b]$. Let f be a nonzero element of $C[a, b]$ that is w -orthogonal to Π_n . Then f changes sign at least $n + 1$ times on (a, b) .

Proof Because $1 \in \Pi_n$, we have $\int_a^b f(x)w(x) dx = 0$, and this shows that f changes sign at least once. Suppose that f changes sign only r times, with $r \leq n$. Choose points t_i so that

$$a = t_0 < t_1 < t_2 < \cdots < t_r < t_{r+1} = b$$

and so that f is of one sign on each interval

$$(t_0, t_1), (t_1, t_2), \dots, (t_r, t_{r+1})$$

The polynomial

$$p(x) = \prod_{i=1}^r (x - t_i)$$

has the same sign property, and therefore $\int_a^b f(x)p(x)w(x) dx \neq 0$. Since $p \in \Pi_n$, this is a contradiction. ■

If the weight function is $w(x) = 1$ and if the interval is $[-1, 1]$, we have the situation originally investigated by Gauss. Here are two of the formulas for this case, corresponding to $n = 1$ and $n = 4$:

$$\int_{-1}^1 f(x) dx \approx f(-\alpha) + f(\alpha) \tag{8}$$

where $\alpha = 1/\sqrt{3}$, and

$$\int_{-1}^1 f(x) dx \approx A_0 f(x_0) + A_1 f(x_1) + \cdots + A_4 f(x_4) \tag{9}$$

where

$$-x_0 = x_4 = \frac{1}{3}\sqrt{5 + 2\sqrt{10/7}} \approx 0.90617\,98459\,38664$$

$$-x_1 = x_3 = \frac{1}{3}\sqrt{5 - 2\sqrt{10/7}} \approx 0.53846\,93101\,05683$$

$$x_2 = 0.0$$

$$A_0 = A_4 = 0.3(0.7 + 5\sqrt{0.7})/(2 + 5\sqrt{0.7}) \approx 0.23692\,68850\,56189$$

$$A_1 = A_3 = 0.3(-0.7 + 5\sqrt{0.7})/(-2 + 5\sqrt{0.7}) \approx 0.47862\,86704\,99366$$

$$A_2 = 128/225 \approx 0.56888\,88888\,88889$$

By using the exact nodes and coefficients, we can compute these numbers to any desired precision. In fact, the nodes x_i are roots of the Legendre polynomials $p_2(x) = x^3 - \frac{1}{3}x$ and $p_5(x) = x^5 - \frac{10}{9}x^3 + \frac{5}{21}x$, respectively, which were determined in Example 2 of Section 6.8 (p. 400).

The nodes x_i and coefficients A_i for many integration formulas of various types are found in handbooks such as Abramowitz and Stegun [1964]. The tables found there cover values of n from 1 to 9 as well as $n = 11, 15, 19, 23, 31, 39, 47, 63, 79, 95$. For well-behaved integrands, modest accuracy is obtained with only a few function evaluations using Gaussian formulas, and even more accuracy can be obtained by using some of the high-order formulas.

A short pseudocode is presented for approximating the integral

$$\int_a^b f(x) dx$$

by incorporating the preceding five-point Gaussian integration formula. Here we change intervals as described in Equation (9) of Section 7.2 (p. 486) and exploit the symmetry in Equation (8):

input

```

 $x_0 \leftarrow 0$ 
 $x_1 \leftarrow 0.53846\,93101\,05683$ 
 $x_2 \leftarrow 0.90617\,98459\,38664$ 
 $A_0 \leftarrow 0.56888\,88888\,88889$ 
 $A_1 \leftarrow 0.47862\,86704\,99366$ 
 $A_2 \leftarrow 0.23692\,68850\,56189$ 

```

```

 $u \leftarrow (a + b)/2$ 

```

```

 $S \leftarrow A_0 f(u)$ 

```

```

for  $i = 1$  to  $2$  do

```

```

     $u \leftarrow ((b - a)x_i + a + b)/2$ 

```

```

     $v \leftarrow ((a - b)x_i + a + b)/2$ 

```

```

     $S \leftarrow S + A_i [f(u) + f(v)]$ 

```

```

end do

```

```

 $S \leftarrow (b - a)S/2$ 

```

```

output  $S$ 

```

The computation of the coefficients A_i in a Gaussian formula proceeds in the manner already indicated for non-Gaussian formulas, once the nodes x_i have been determined. The nodes are, in turn, the roots of a certain polynomial. This polynomial depends on $n + 1$, and we now denote it by q_{n+1} . The polynomial q_{n+1} is uniquely defined by two conditions:

1. q_{n+1} is a monic polynomial of degree $n + 1$.
2. q_{n+1} is w -orthogonal to Π_n .

The word **monic** means that in q_{n+1} , the coefficient of x^{n+1} is unity. The condition of **w -orthogonality** is that

$$\int_a^b q_{n+1}(x)p(x)w(x) dx = 0 \quad \text{for all } p \in \Pi_n$$

These so-called **orthogonal polynomials** are useful in various branches of mathematics, and much is known about them. They can be generated by a recursive algorithm, as discussed in Section 6.8 (p. 401).

EXAMPLE 1 Find the Gaussian quadrature rule when $[a, b] = [-1, 1]$, $w(x) = 1$, and $n = 2$.

Solution From Theorem 5 in Section 6.8 (p. 400), the orthogonal polynomials can be determined recursively, as was done in Example 2 in that section. Hence,

$$\begin{aligned} q_0(x) &= 1 \\ q_1(x) &= x \\ q_2(x) &= x^2 - \frac{1}{3} \\ q_3(x) &= x^3 - \frac{3}{5}x \end{aligned}$$

The roots 0 and $\pm\sqrt{3/5}$ of q_3 are the nodes in the quadrature formula

$$\int_{-1}^1 f(x) \, dx \approx \frac{5}{9}f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9}f(0) + \frac{5}{9}f\left(\sqrt{\frac{3}{5}}\right)$$

The constants, $5/9$ and $8/9$, can be found by the method of undetermined coefficients. (See Problem 7.3.21, p. 500.)

■

Convergence and Error Analysis

Some of the favorable characteristics of Gaussian quadrature formulas will now be discussed.

LEMMA 1 Lemma on Gaussian Quadrature Formula

In a Gaussian quadrature formula, the coefficients are positive, and moreover, their sum is $\int_a^b w(x) \, dx$.

Proof Fix n , and let q be a polynomial of degree $n + 1$ that is w -orthogonal to Π_n , as in Theorem 1. The zeros of q are denoted by $x_0, x_1, \dots, x_n$, and these are the nodes in the Gaussian formula (5). Let p be the polynomial $q(x)/(x - x_j)$ for some fixed j . Since p^2 is of degree at most $2n$, the Gaussian formula will be exact for it. Consequently,

$$0 < \int_a^b p^2(x)w(x) \, dx = \sum_{i=0}^n A_i p^2(x_i) = A_j p^2(x_j)$$

From this we conclude that $A_j > 0$. Because the Gaussian formula is exact for $f(x) \equiv 1$,

$$\int_a^b w(x) \, dx = \sum_{i=0}^n A_i$$

■

A remarkable theorem of Stieltjes establishes the convergence of the Gaussian formulas when $n \rightarrow \infty$. Fixing w and $[a, b]$, we have a Gaussian quadrature formula for each $n \in \{0, 1, 2, \dots\}$. These we write now as in the following theorem.

THEOREM 3 Theorem on Gaussian Quadrature Convergence

If f is continuous on $[a, b]$, then the approximations in

$$\int_a^b f(x)w(x) dx \approx \sum_{i=0}^n A_{ni} f(x_{ni}) \quad (n \geq 0) \quad (10)$$

converge to the integral as $n \rightarrow \infty$.

Proof Let $\varepsilon > 0$. By the Weierstrass Approximation Theorem (Theorem 8 of Section 6.1, p. 320), a polynomial p exists for which $|f(x) - p(x)| < \varepsilon$ on $[a, b]$. For any n such that $2n$ exceeds the degree of p , the n th Gaussian formula will integrate p correctly. Using elementary inequalities and Lemma 1 on Gaussian Quadrature Formula, we have

$$\begin{aligned} & \left| \int_a^b f(x)w(x) dx - \sum_{i=0}^n A_{ni} f(x_{ni}) \right| \\ & \leq \left| \int_a^b f(x)w(x) dx - \int_a^b p(x)w(x) dx \right| + \left| \sum_{i=0}^n A_{ni} p(x_{ni}) - \sum_{i=0}^n A_{ni} f(x_{ni}) \right| \\ & \leq \int_a^b |f(x) - p(x)|w(x) dx + \sum_{i=0}^n A_{ni} |p(x_{ni}) - f(x_{ni})| \\ & \leq \varepsilon \int_a^b w(x) dx + \varepsilon \sum_{i=0}^n A_{ni} = 2\varepsilon \int_a^b w(x) dx \end{aligned} \quad \blacksquare$$

THEOREM 4 Theorem on Gaussian Formula with Error Term

Consider a Gaussian formula with error term:

$$\int_a^b f(x)w(x) dx = \sum_{i=0}^{n-1} A_i f(x_i) + E$$

For an $f \in C^{2n}[a, b]$, we have

$$E = \frac{f^{(2n)}(\xi)}{(2n)!} \int_a^b q^2(x)w(x) dx$$

where $a < \xi < b$ and $q(x) = \prod_{i=0}^{n-1} (x - x_i)$.

Proof By Hermite interpolation (Section 6.3, p. 338), a polynomial p of degree at most $2n - 1$ exists such that

$$p(x_i) = f(x_i) \quad p'(x_i) = f'(x_i) \quad (0 \leq i \leq n - 1)$$

The error formula for this interpolation is

$$f(x) - p(x) = f^{(2n)}(\zeta(x))q^2(x)/(2n)! \tag{11}$$

as given in Theorem 2 of Section 6.3 (p. 344). It follows that

$$\int_a^b f(x)w(x) \, dx - \int_a^b p(x)w(x) \, dx = \frac{1}{(2n)!} \int_a^b f^{(2n)}(\zeta(x))q^2(x)w(x) \, dx$$

Now use the fact that p is of degree at most $2n - 1$ to see that

$$\int_a^b p(x)w(x) \, dx = \sum_{i=0}^{n-1} A_i p(x_i) = \sum_{i=0}^{n-1} A_i f(x_i)$$

Furthermore, the Mean-Value Theorem for Integrals can be used to write

$$\int_a^b f^{(2n)}(\zeta(x))q^2(x)w(x) \, dx = f^{(2n)}(\xi) \int_a^b q^2(x)w(x) \, dx$$

This requires continuity of $f^{(2n)}(\zeta(x))$, which can be inferred from Equation (11). The desired error formula results from making the substitution described. ■

Further references for Gaussian quadrature are Krylov [1962], Davis and Rabinowitz [1956, 1984], Stroud and Secrest [1966], and Abramowitz and Stegun [1956, 1964].

PROBLEMS 7.3

- 1. In Equation (2), let $n = 1$, $a = 0$, and $b = 1$. Find all cases of the formula that are exact for $f \in \Pi_3$.
- 2. (Continuation) Solve the preceding problem with $n = 2$.
- 3. Derive the following Gaussian quadrature rule:

$$\int_{-1}^1 f(x) \, dx \approx A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + A_3 f(x_3)$$

where

$$A_0 = A_2 = \frac{1}{2} \left(1 + \frac{1}{6} \sqrt{10/3} \right) \approx 0.34785 \, 48451 \, 37454$$

$$A_1 = A_3 = \frac{1}{2} \left(1 - \frac{1}{6} \sqrt{10/3} \right) \approx 0.65214 \, 51548 \, 62546$$

$$-x_0 = x_2 = \sqrt{\frac{1}{7} (3 - 4\sqrt{0.3})} \approx 0.86113 \, 63115 \, 94052$$

$$-x_1 = x_3 = \sqrt{\frac{1}{7} (3 + 4\sqrt{0.3})} \approx 0.33998 \, 10435 \, 848456$$

- 4. Verify the correctness of Equation (8) as a Gaussian formula.
- 5. Verify that the nodes in Equation (9) are correct.

6. We can determine a polynomial q of degree $n + 1$ that is orthogonal to Π_n by writing

$$q(x) = x^{n+1} + c_1x^n + \cdots + c_{n+1}$$

and imposing the conditions $\int_a^b q(x)x^k w(x) dx = 0$ for $0 \leq k \leq n$. The resulting system of $n + 1$ equations in the $n + 1$ unknowns $c_1, c_2, \dots, c_{n+1}$ can then be solved. Carry out this process to obtain q_5 as needed in Problem 7.3.3 (p. 498). Do you think that this is a good way to obtain q ?

7. a. Find a formula of the form

$$\int_0^1 xf(x) dx \approx \sum_{i=0}^n A_i f(x_i)$$

with $n = 1$, that is exact for all polynomials of degree 3.

b. Repeat with $n = 2$, making the formula exact on Π_5 .

8. a. Determine appropriate values of A_i and x_i so that the quadrature formula

$$\int_{-1}^1 x^2 f(x) dx \approx \sum_{i=0}^n A_i f(x_i)$$

will be correct when f is any polynomial of degree 3. Use $n = 1$.

b. Repeat when f is any polynomial of degree 5, using $n = 2$.

9. Find a quadrature formula

$$\int_{-1}^1 f(x) dx \approx c \sum_{i=0}^2 f(x_i)$$

that is exact for all quadratic polynomials.

10. a. If the integration formula

$$\int_{-1}^1 f(x) dx \approx f(\alpha) + f(-\alpha)$$

is to be exact for all quadratic polynomials, what value of α should be used? Answer the same question for all cubic polynomials.

Repeat Part a for polynomials of the forms:

b. $f(x) = a + bx + cx^3 + dx^4$

c. $f(x) = a + \sum_{i=1}^n b_i x^{2i-1} + cx^{2n}$

11. For what value of α is this formula exact on Π_3 ?

$$\int_0^2 f(x) dx \approx f(\alpha) + f(2 - \alpha)$$

12. Prove that if the interval is symmetric with respect to the origin and if w is an even function, then the Gaussian nodes will be symmetric and $A_i = A_{n-i}$ for $0 \leq i \leq n$.

13. Prove that every quadrature formula of the type

$$\int_a^b f(x) dx \approx \sum_{i=0}^n A_i f(x_i)$$

is exact on some infinite-dimensional subspace of $C[a, b]$.

14. Prove the following: If

$$\int_a^b f(x)w(x)dx = \sum_{i=0}^n A_i f(x_i)$$

for all $f \in \Pi_{2n+1}$, then the polynomial $\prod_{i=0}^n (x - x_i)$ is orthogonal to Π_n on $[a, b]$ with respect to w .

15. Determine the coefficients
- A_0
- ,
- A_1
- , and
- A_2
- that make the formula

$$\int_0^2 f(x)dx \approx A_0 f(0) + A_1 f(1) + A_2 f(2)$$

exact for all polynomials of degree 3.

16. Using only $f(0)$, $f'(-1)$, and $f''(1)$, compute an approximation to $\int_{-1}^1 f(x)dx$ that is exact for all quadratic polynomials. Is the approximation exact for polynomials of degree 3? Why or why not?
17. If the formula

$$\int_0^1 xf(x)dx = \sum_{i=0}^4 A_i f(x_i)$$

is correct for all $f \in \Pi_9$ (ninth-degree polynomials), then $x_0, x_1, \dots, x_4$ must be the zeros of a fifth-degree polynomial q that has what properties?

18. If the formula

$$\int_1^2 (x^4 - 1)f(x)dx = Af(x_0) + Bf(x_1) + Cf(x_2)$$

is correct for all f that are polynomials of degree ≤ 5 , then x_0, x_1 , and x_2 must be roots of a polynomial q having what properties?

19. Which of these polynomials is orthogonal to Π_2 on the interval $[0, 1]$ with weight function $w(x) = 1$? $1 + x$, $x - \frac{1}{2}$, $x^2 - 3x + 1$, $35x^4 - 60x^2 + 32x - 3$, $x^3 - 3x^2 + x - 1$.
20. Find a nonzero polynomial that is orthogonal to Π_2 on the interval $[-1, 1]$ with respect to the weight function $1 + x^2$.
21. Consider a numerical integration rule of the form

$$\int_{-1}^1 f(x)dx \approx Af\left(-\sqrt{\frac{3}{5}}\right) + Bf(0) + Cf\left(\sqrt{\frac{3}{5}}\right)$$

- a. What is the linear system that must be solved in the method of undetermined coefficients for finding A , B , and C ? Solve for A , B , and C .
- b. What three integrals must be evaluated to determine A , B , and C in a Newton-Cotes formula? Solve for A , B , and C .
22. Show how the Gaussian quadrature rule

$$\int_{-1}^1 f(x)dx \approx \frac{5}{9}f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9}f(0) + \frac{5}{9}f\left(\sqrt{\frac{3}{5}}\right)$$

can be used for $\int_a^b f(x)dx$. Apply this result to evaluate:

a. $\int_0^{\pi/2} x dx$

b. $\int_0^4 \frac{\sin t}{t} dt$

23. Given $f(0)$, $f'(-1)$, and $f''(1)$, compute an approximation to $\int_{-1}^1 x^2 f(x) dx$ by the method of undetermined coefficients. Your formula should give exact results for all f in Π_2 .

24. Find A , B , and C so that the numerical integration rule of the form

$$\int_{-1}^1 x f(x) dx \approx Af(-1) + Bf(0) + Cf(+1)$$

is exact for all polynomials of maximum degree m . What is m ?

25. Using the method of undetermined coefficients, find A , B , and C in the following rule, which should give exact results for polynomials of degree 2:

$$\int_{-3h}^h f(x) dx \approx h[Af(0) + Bf(-h) + Cf(-2h)]$$

26. (Continuation) How can a numerical integration rule of the form given in the preceding problem be used to obtain a rule for solving an ordinary differential equation of the following form?

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases}$$

27. Find the coefficients and nodes of a Gaussian quadrature formula of the form

$$\int_0^1 x^4 f(x) dx \approx A_0 f(x_0) + A_1 f(x_1)$$

28. Prove that no Gaussian quadrature formula with n nodes can be exact on Π_{2n} .
29. What happens in the theory of Gaussian quadrature if the function w is not positive?
30. Use information from Section 7.2 (p. 484) to find the Gaussian formulas pertaining to the interval $[-1, 1]$ and the weight function $w(x) = \sqrt{1-x^2}$. *Hint:* Problems 7.2.26–27 (p. 491) may be helpful.
31. Determine the nodes and weights for the Gaussian formula of the form

$$\int_{-1}^1 x^4 f(x) dx \approx A_0 f(x_0) + A_1 f(x_1)$$

32. Derive the exact values for the nodes and weights for each of the following formulas. Also determine the highest degree polynomial for which the formula is exact.
- Equation (3)
 - Equation (8)
 - Equation (9)
33. Determine the composite rule for $\int_a^b f(x) dx$ using
- two-point Gaussian formula
 - three-point Gaussian formula
34. Is Theorem 1, on Gaussian quadrature, true if we drop the hypothesis about the degree of q ? Retain the hypothesis that each point $x_0, x_1, \dots, x_n$ is a zero of q , but additional zeros may exist.

COMPUTER PROBLEM 7.3

- Write a computer program to test whether Hermite's quadrature formula, Equation (4), with $n = 4$ is exact for all functions $p(x)(1-x^2)^{-1/2}$, where $p \in \Pi_9$. It suffices to test the formula for the ten test functions $T_k(x)(1-x^2)^{-1/2}$, where T_k is the k th Chebyshev polynomial defined by $T_k(x) = \cos(k \cos^{-1} x)$.

7.4 Romberg Integration

The algorithm to be described now goes by the name of Romberg, who first gave a recursive form for the method. We suppose that an approximate value is required for the integral

$$I = \int_a^b f(x) dx \quad (1)$$

The function f and the interval $[a, b]$ will remain fixed in the discussion.

Recursive Trapezoid Rule

Let the trapezoid rule for I using n subintervals of width $h = (b - a)/n$ be denoted by $T(n)$. From Equation (4) in Section 7.2 (p. 482), we have

$$T(n) = h \sum_{i=0}^n{}'' f(a + ih) = \frac{(b - a)}{n} \sum_{i=0}^n{}'' f\left(a + i \frac{(b - a)}{n}\right) \quad (2)$$

Here, the double prime on the summation sign signifies that the first and last terms are to be *halved*.

EXAMPLE 1 What are the explicit formulas for $T(1)$, $T(2)$, $T(4)$, and $T(8)$ in the case in which the interval is $[0, 1]$?

Solution Using Equation (2), we have

$$T(1) = \frac{1}{2} f(0) + \frac{1}{2} f(1)$$

$$T(2) = \frac{1}{4} f(0) + \frac{1}{2} \left[f\left(\frac{1}{2}\right) \right] + \frac{1}{4} f(1)$$

$$T(4) = \frac{1}{8} f(0) + \frac{1}{4} \left[f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right) \right] + \frac{1}{8} f(1)$$

$$T(8) = \frac{1}{16} f(0) + \frac{1}{8} \left[f\left(\frac{1}{8}\right) + f\left(\frac{1}{4}\right) + f\left(\frac{3}{8}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{5}{8}\right) + f\left(\frac{3}{4}\right) + f\left(\frac{7}{8}\right) \right] + \frac{1}{16} f(1) \quad \blacksquare$$

It is clear that if $T(2n)$ is to be computed, then we can take advantage of the work already done in the computation of $T(n)$. Only the terms present in $T(2n)$ but *not already present* in $T(n)$ must be computed. For example, from the preceding equations, we see that

$$T(2) = \frac{1}{2} T(1) + \frac{1}{2} \left[f\left(\frac{1}{2}\right) \right]$$

$$T(4) = \frac{1}{2} T(2) + \frac{1}{4} \left[f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) \right]$$

$$T(8) = \frac{1}{2} T(4) + \frac{1}{8} \left[f\left(\frac{1}{8}\right) + f\left(\frac{3}{8}\right) + f\left(\frac{5}{8}\right) + f\left(\frac{7}{8}\right) \right]$$

With $h = (b - a)/(2n)$, the general formula pertaining to any interval $[a, b]$ is as follows:

$$T(2n) = \frac{1}{2} T(n) + h \left[f(a + h) + f(a + 3h) + f(a + 5h) + \cdots + f(a + (2n - 1)h) \right]$$

or

$$T(2n) = \frac{1}{2}T(n) + h \sum_{i=1}^n f(a + (2i-1)h) \quad (3)$$

We prove Equation (3) by first writing $T(2n)$ in terms of $T(n)$:

$$T(2n) = \frac{1}{2}T(n) + \left[T(2n) - \frac{1}{2}T(n) \right]$$

By using Equation (2), we can rewrite the bracketed expression in the form

$$T(2n) - \frac{1}{2}T(n) = h \sum_{i=0}^{2n} f(a + ih) - h \sum_{i=1}^{2n} f(a + 2ih) = h \sum_{i=1}^n f(a + (2i-1)h)$$

Because each term in the second summation is of the form $f(a + 2ih)$, the second summation cancels all even-numbered terms in the first summation. In the resulting sum, the range of i is obtained by noting that in the first summation the first odd term is $f(a + ih)$ so we want $2i - 1 = 1$, and the last odd term is $f(a + (2n-1)h)$ so we want $2i - 1 = 2n - 1$. Hence, Equation (3) is established. The integral I , Equation (1), now can be approximated by a recursive trapezoid rule. If there are 2^n uniform subintervals, then Equation (3) provides a recursive trapezoid rule:

$$T(2^n) = \frac{1}{2}T(2^{n-1}) + h_n \sum_{i=1}^{2^{n-1}} f(a + (2i-1)h_n)$$

where

$$h_0 = b - a \quad h_n = h_{n-1}/2 \quad (n \geq 1)$$

Romberg Algorithm

This formula is used in the Romberg algorithm. Letting $R(n, 0)$ denote the trapezoid estimate with 2^n subintervals, we have

$$\begin{cases} R(0, 0) = \frac{1}{2}(b-a)[f(a) + f(b)] \\ R(n, 0) = \frac{1}{2}R(n-1, 0) + h_n \sum_{i=1}^{2^{n-1}} f(a + (2i-1)h_n) \end{cases} \quad (4)$$

The estimates $R(0, 0), R(1, 0), R(2, 0), \dots, R(M, 0)$ are computed for a modest value of M , and there are no duplicate function evaluations. In the remainder of the Romberg algorithm, additional quantities $R(n, m)$ are to be computed. All of these can be interpreted as estimates of the integral I . Further evaluations of the integrand f are not necessary after the element $R(M, 0)$ has been computed. The subsequent columns of the R -array for $n \geq 1$ and $m \geq 1$ are constructed from the formula

$$R(n, m) = R(n, m - 1) + \frac{1}{4^m - 1} [R(n, m - 1) - R(n - 1, m - 1)] \quad (5)$$

This calculation is very simple. It is used to provide a final array of the form

$$\begin{array}{cccccc} R(0, 0) & & & & & \\ R(1, 0) & R(1, 1) & & & & \\ R(2, 0) & R(2, 1) & R(2, 2) & & & \\ R(3, 0) & R(3, 1) & R(3, 2) & R(3, 3) & & \\ R(4, 0) & R(4, 1) & R(4, 2) & R(4, 3) & R(4, 4) & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ R(M, 0) & R(M, 1) & R(M, 2) & R(M, 3) & R(M, 4) & \cdots R(M, M) \end{array}$$

Here is the pseudocode for the Romberg algorithm computed row-wise:

```
input a, b, M
h ← b - a
R(0, 0) ← 1/2(b - a)[f(a) + f(b)]
for n = 1 to M do
    h ← h/2
    R(n, 0) ← 1/2 R(n - 1, 0) + h ∑i=12n-1 f(a + (2i - 1)h)
    for m = 1 to n do
        R(n, m) ← R(n, m - 1) + [R(n, m - 1) - R(n - 1, m - 1)]/(4m - 1)
    end do
end do
output R(n, m) (0 ≤ n ≤ M, 0 ≤ m ≤ n)
```

Usually, only a moderate value of M is needed because the number of function values required is $2^M + 1$. A more sophisticated algorithm might include a procedure for automatically stopping the calculations when a certain prescribed error criterion is met.

Analysis

To explain the origin of Equation (5), we start with the **Euler-Maclaurin formula**:

$$\begin{aligned} \int_0^1 f(t) dt &= \frac{1}{2} [f(0) + f(1)] \\ &+ \sum_{k=1}^{m-1} A_{2k} [f^{(2k-1)}(0) - f^{(2k-1)}(1)] \\ &- A_{2m} f^{(2m)}(\xi_0) \end{aligned} \quad (6)$$

where ξ_0 is some point between 0 and 1. It can be shown that Equation (6) is valid for any function $f \in C^{2m}[0, 1]$. A proof of this important formula is given in Section 7.7 (p. 522). The constants $k! A_k$ are known as **Bernoulli numbers**. They can be

defined by the equation

$$\frac{x}{e^x - 1} = \sum_{k=0}^{\infty} A_k x^k$$

From the basic formula (6), we can obtain the following one in two steps: Apply Equation (6) to the function g defined by $g(t) = f(x_i + ht)$ with $h = x_{i+1} - x_i$ and, in the resulting integral, make the change of variable $t = (x - x_i)/h$:

$$\begin{aligned} \int_{x_i}^{x_{i+1}} f(x) dx &= \frac{h}{2} [f(x_i) + f(x_{i+1})] \\ &+ \sum_{k=1}^{m-1} A_{2k} h^{2k} [f^{(2k-1)}(x_i) - f^{(2k-1)}(x_{i+1})] \\ &- A_{2m} h^{2m+1} f^{(2m)}(\xi_i) \end{aligned} \quad (7)$$

In Equation (7), we apply the operation $\sum_{i=0}^{2^n-1}$ to both sides. If $x_i = a + ih$ for $0 \leq i \leq 2^n$ and $h = (b - a)/2^n$, then

$$\begin{aligned} \int_a^b f(x) dx &= \frac{h}{2} \sum_{i=0}^{2^n-1} [f(x_i) + f(x_{i+1})] \\ &+ \sum_{k=1}^{m-1} A_{2k} h^{2k} [f^{(2k-1)}(a) - f^{(2k-1)}(b)] \\ &- A_{2m} (b - a) h^{2m} f^{(2m)}(\xi) \end{aligned} \quad (8)$$

for some $\xi \in (a, b)$. In obtaining this equation, the error term has been handled as described in Problem 7.4.2 (p. 506). Equation (8) is interpreted as follows: The first term on the right is the trapezoidal estimate of I , Equation (1), using subintervals of size $h = (b - a)/2^n$. Thus, Equation (8) tells us that

$$I = R(n, 0) + c_2 h^2 + c_4 h^4 + c_6 h^6 + \cdots + c_{2m-2} h^{2m-2} + c_{2m} h^{2m} f^{(2m)}(\xi) \quad (9)$$

where $f \in C^{2m}[a, b]$ and $\xi \in (a, b)$. This equation is valid for arbitrary h , and the coefficients $c_2, c_4, \dots, c_{2m}$ are independent of h . Since Equation (9) is a special case of Equation (20) of Section 7.1 (p. 475), we immediately have Equation (4) by the Richardson extrapolation analysis in Section 7.1 (p. 471).

To take full advantage of the Romberg algorithm, we shall want the function f to belong to class $C^{2m}[a, b]$, with as high a value of m as possible. This hypothesis justifies the use of the Euler-Maclaurin formula and implies that the quantities $R(n, m)$ converge to the integral of f with an error that is $\mathcal{O}(h^{2m})$, where $h = (b - a)2^{-n}$. What happens if f is assumed to be only continuous?

THEOREM 1 Theorem on Convergence of the Romberg Algorithm

If $f \in C[a, b]$, then each column in the Romberg array converges to the integral of f . Thus, for each m ,

$$\lim_{n \rightarrow \infty} R(n, m) = \int_a^b f(x) dx$$

Proof We begin with the first column, which contains trapezoidal estimates of the integral, I . The trapezoid rule with k subintervals can be written in the form

$$h \sum_{i=0}^k f(a + ih) = \frac{1}{2}h \sum_{i=0}^{k-1} f(a + ih) + \frac{1}{2}h \sum_{i=1}^k f(a + ih)$$

The right side of this equation represents the average of two Riemann sums for I . Because $h = (b - a)/k$, the maximum width of subintervals converges to 0 as $k \rightarrow \infty$. Hence, by the theory of the Riemann integral, both Riemann sums converge to I . Of course, their average also converges to I . This proves that $\lim_{n \rightarrow \infty} R(n, 0) = I$. As for the second column, we note that

$$R(n, 1) = \frac{4}{3}R(n, 0) - \frac{1}{3}R(n - 1, 0)$$

from which we get

$$\lim_{n \rightarrow \infty} R(n, 1) = \frac{4}{3}I - \frac{1}{3}I = I$$

All subsequent columns can be analyzed in the same way. ■

PROBLEMS 7.4

- Derive Equation (7) starting with Equation (6).
- Derive Equation (8) from Equation (7), and in particular, justify the conversion from

$$h^{2m+1} \sum_{i=0}^{2^n-1} f^{(2m)}(\xi_i) \quad \text{to} \quad (b-a)h^{2m} f^{(2m)}(\xi)$$

- Establish the following equation in which $h = 1/2^n$:

$$I = \frac{4}{3}T\left(f, \frac{h}{2}\right) - \frac{1}{3}T(f, h) - \sum_{n=1}^{\infty} \frac{4^n - 1}{3(4^n)} c_{2n+2} h^{2n+2}$$

where

$$I = \int_0^1 f(x) dx \quad \text{and} \quad T(f, h) = h \sum_{i=0}^{2^n} f(ih)$$

- Show that the second column in the Romberg array is the result of using Simpson's rule on f . (See Equation (6) in Section 7.2, p. 483.)
- Prove by induction that

$$I - R(n, m - 1) = ah^{2m} + bh^{2m+2} + ch^{2m+4} + \dots$$

- Apply the Romberg algorithm to find $R(2, 2)$ for these integrals:

a. $\int_1^3 \frac{dx}{x}$

b. $\int_0^{\pi/2} \left(\frac{x}{\pi}\right)^2 dx$ (in terms of π)

- Suppose that $S(f, h)$ is a quadrature rule for the integral I in Equation (1) and that the error series is $c_4h^4 + c_6h^6 + \dots$. Combine $S(f, h)$ with $S(f, h/3)$ to find a more accurate approximation to I .

8. In the Romberg algorithm, $R(n, 0)$ is an estimate of $\int_a^b f(x) dx$ using the trapezoid rule with 2^n subintervals. How many evaluations of $f(x)$ are needed to compute $R(i, j)$ for $0 \leq i \leq N$ and $0 \leq j \leq N$?
9. If the trapezoid rule satisfied the equation

$$\int_a^b f(x) dx = T(f, h) + c_1 h + c_2 h^2 + c_3 h^3 + \cdots$$

instead of Equation (9), then how would we have to modify Formula (5)?

10. In the Romberg algorithm, the elements in the second column satisfy

$$R(i, 1) = I + C_4 h_i^4 + C_6 h_i^6 + \cdots$$

where $I = \int_a^b f(x) dx$ and $h_i = (b - a)/2^i$. Derive the formula for computing elements in the third column and the first term in its error series.

11. (**Milne's rule**) Express $R(2, 2)$ in terms of elements in the first column of the Romberg array. Show that $R(3, 3)$ is *not* a Newton-Cotes formula but that $R(2, 2)$ is.
12. Show that Equation (3) follows immediately from the fact that

$$\sum_{0 \leq i \leq 2n} f(a + ih) - \sum_{\substack{0 \leq i \leq 2n \\ i \text{ even}}} f(a + ih) = \sum_{\substack{0 \leq i \leq 2n \\ i \text{ odd}}} f(a + ih)$$

COMPUTER
PROBLEM 7.4

1. Write a subprogram to carry out the Romberg algorithm for a function f defined on an arbitrary interval $[a, b]$. The user will specify the number of rows to be computed in the array and will want to see the entire array when it has been computed. Write a main program and test your Romberg subprogram on these three examples:

- a. $\int_0^1 \frac{\sin x}{x} dx$
- b. $\int_{-1}^1 \frac{\cos x - e^x}{\sin x} dx$
- c. $\int_1^\infty (xe^x)^{-1} dx$

The routines for these integrals should be written to avoid serious loss of significance due to subtraction. Also, it is customary to define a function f at any questionable point x_0 by the equation $f(x_0) = \lim_{x \rightarrow x_0} f(x)$. If the limit exists, this method guarantees continuity of f at x_0 . For the third example, make a suitable change of variable, such as $x = 1/t$. Compute seven rows in the Romberg array. Print the array in each case with a format that enables the convergence to be observed.

7.5 Adaptive Quadrature

Adaptive quadrature methods are intended to compute definite integrals by automatically taking into account the behavior of the integrand. Ideally, the user supplies only the integrand f , the interval $[a, b]$, and the accuracy ε desired for computing

the integral

$$\int_a^b f(x) dx \quad (1)$$

The program then divides the interval into pieces of varying length so that numerical integration on the subintervals will produce results of acceptable precision.

Here a typical adaptive quadrature method will be developed. It will use Simpson's rule on subintervals together with an estimate of the errors involved. **Simpson's rule** is given by Equation (6) in Section 7.2 (p. 483):

$$\begin{aligned} \int_a^b f(x) dx &= S(a, b) - \frac{1}{90}[(b-a)/2]^5 f^{(4)}(\xi) \\ S(a, b) &= \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] \end{aligned} \quad (2)$$

for some $\xi \in (a, b)$. The main idea is that if Simpson's rule on a given subinterval is not sufficiently accurate, that interval will be divided into two equal parts, and Simpson's rule will be used on each half. This procedure will be repeated in an effort to obtain an approximation to the integral with the same accuracy over all subintervals involved. At the end, we shall have computed the integral with n applications of Simpson's rule,

$$\int_a^b f(x) dx = \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x) dx = \sum_{i=1}^n (S_i + e_i) = \sum_{i=1}^n S_i + \sum_{i=1}^n e_i$$

where S_i is an approximation to the integral on the interval $[x_{i-1}, x_i]$ and e_i is the associated *local* error. If

$$|e_i| \leq \varepsilon(x_i - x_{i-1})/(b-a) \quad (3)$$

then the *total* error will be bounded by

$$\left| \sum_{i=1}^n e_i \right| \leq \sum_{i=1}^n |e_i| \leq \frac{\varepsilon}{b-a} \sum_{i=1}^n (x_i - x_{i-1}) = \varepsilon$$

Thus, the *local* error criterion (3) leads to an *absolute* error bound

$$\left| \int_a^b f(x) dx - \sum_{i=1}^n S_i \right| \leq \varepsilon$$

From Equation (2), the *basic* Simpson's rule on the interval $[u, v]$ is described by

$$\int_u^v f(x) dx = S(u, v) - \frac{1}{90}[(v-u)/2]^5 f^{(4)}(\xi_1) \quad (4)$$

for some $\xi_1 \in (u, v)$. If the interval of integration is divided into two equal subintervals at the midpoint $w = (u + v)/2$, then a more accurate value of the integral can be computed by using Simpson's rule on each subinterval. Here is the result of doing so:

$$\begin{aligned}
 \int_u^v f(x) dx &= \int_u^w f(x) dx + \int_w^v f(x) dx \\
 &= S(u, w) - \frac{1}{90}[(w - u)/2]^5 f^{(4)}(\xi_2) + S(w, v) \\
 &\quad - \frac{1}{90}[(v - w)/2]^5 f^{(4)}(\xi_3) \\
 &= S^* + S^{**} - \frac{1}{90} \left(\frac{v - u}{2^2} \right)^5 [f^{(4)}(\xi_2) + f^{(4)}(\xi_3)] \\
 &= S^* + S^{**} - \frac{1}{2^9} \cdot \frac{1}{90} (v - u)^5 f^{(4)}(\xi)
 \end{aligned} \tag{5}$$

In this calculation, we have set

$$\begin{aligned}
 S^* &\equiv S(u, w) & S^{**} &\equiv S(w, v) \\
 f^{(4)}(\xi) &\equiv \frac{1}{2} [f^{(4)}(\xi_2) + f^{(4)}(\xi_3)]
 \end{aligned} \tag{6}$$

where $\xi_2 \in (u, w)$, $\xi_3 \in (w, v)$, and $\xi \in (u, v)$. Equation (6) is justified if we assume $f^{(4)}$ is continuous. As is usual in such formulas, the error term cannot be estimated or bounded unless some knowledge of $f^{(4)}$ is available. For automatic computation, however, it is imperative to find a way of estimating the magnitude of $f^{(4)}(\xi)$. This will be accomplished by assuming that over small intervals, $f^{(4)}$ is *constant*. In particular, let us assume that $f^{(4)}(\xi_1) = f^{(4)}(\xi)$ in Equations (4) and (5). Then the terms involving $f^{(4)}$ can be eliminated by multiplying Equation (5) by 16/15 and subtracting 1/15 times Equation (4). The result is

$$\int_u^v f(x) dx \approx S^* + S^{**} + \frac{1}{15} [S^* + S^{**} - S(u, v)] \tag{7}$$

The groundwork has now been laid for an adaptive quadrature method. Suppose that the integral (1) is to be computed numerically with a prescribed tolerance ε . If $f^{(4)}$ exists and is slowly varying, then the approximation in Equation (7) will be satisfactory on small intervals. The adaptive algorithm starts by considering the interval $[a, b]$. For this interval (as well as for others that are considered subsequently), a vector having six components is constructed:

$$[a, h, f(a), f(a + h), f(a + 2h), S] \quad h = \frac{1}{2}(b - a) \tag{8}$$

This vector contains the Simpson estimate $S = S(a, b)$, as well as the data needed to compute it by the equation

$$S = \frac{h}{3} [f(a) + 4f(a + h) + f(a + 2h)]$$

Now we compute $c = a + h$, $S^* = S(a, c)$, and $S^{**} = S(c, b)$. As explained previously, $S^* + S^{**}$ is a more refined estimate of the integral. To see whether it is good enough (that is, whether it meets the tolerance ε), we shall use Equations (7) and (3). Hence, we test the inequality

$$|S^* + S^{**} - S|/15 < \varepsilon(2h)/(b - a) \quad (9)$$

If this is true, $S^* + S^{**} + [S^* + S^{**} - S]/15$ is accepted as the value of the integral (1), in accordance with Equation (7). If Inequality (9) fails, the interval is subdivided into two subintervals of equal length; namely, $[a, c]$ and $[c, b]$, where $c = (a + b)/2$. For each of these intervals, we construct vectors as described previously. The vector in (8) is discarded, and in its place we have two new vectors:

$$\begin{aligned} [a, h/2, f(a), f(y), f(c), S^*] & \quad y = a + h/2 \\ [c, h/2, f(c), f(z), f(b), S^{**}] & \quad z = c + h/2 \end{aligned}$$

Notice that only two new function evaluations, $f(y)$ and $f(z)$, are needed to compute $S^* = S(a, c)$ and $S^{**} = S(c, b)$ because all other quantities have already been computed, and they are now being stored in a particular format. The process that was applied to the vector in (8) will eventually be applied to every vector generated by the algorithm.

A description of the general step in the algorithm goes as follows: At any stage, the sum of integrals of f over certain subintervals will accumulate in a variable Σ . At the same time, a *workstack* of vectors described earlier exists, each of which corresponds to an interval on which the integral of f has yet to be computed satisfactorily. One of these vectors, say

$$[u, h, \bar{u}, \bar{w}, \bar{v}, S]$$

is taken off the stack. It will have these properties:

$$\begin{aligned} w &= u + h & v &= u + 2h \\ \bar{u} &= f(u) & \bar{w} &= f(w) & \bar{v} &= f(v) \\ S &= (\bar{u} + 4\bar{w} + \bar{v})h/3 \end{aligned}$$

Next, we replace h by $h/2$ and compute

$$\begin{aligned} y &= u + h & z &= u + 3h \\ \bar{y} &= f(y) & \bar{z} &= f(z) \\ S^* &= (\bar{u} + 4\bar{y} + \bar{w})h/3 \\ S^{**} &= (\bar{w} + 4\bar{z} + \bar{v})h/3 \end{aligned}$$

We wish to know whether $S^* + S^{**}$, as an estimate of $\int_u^v f(x) dx$, passes the error test, which is

$$|S^* + S^{**} - S| \leq 30\varepsilon h/(b - a) \quad (10)$$

If it passes the test, then $S^* + S^{**} + (S^* + S^{**} - S)/15$ is added to Σ , and a new vector is taken from the stack for processing. If the test fails, then two new vectors are added to the stack:

$$[u, h, \bar{u}, \bar{y}, \bar{w}, S^*]$$

$$[u + 2h, h, \bar{w}, \bar{z}, \bar{v}, S^{**}]$$

At this stage, another vector is removed from the stack and processed as before. The size of the stack is not allowed to exceed n vectors, where n is a parameter set by the user. This helps to prevent the algorithm from becoming infinite.

A pseudocode for this algorithm is given next. It follows the description just given. Some care is taken to save quantities that may be needed later. The vectors in the workstack are denoted by $v^{(1)}$, $v^{(2)}$, and so on. Each vector has six components:

$$v^{(k)} = [v_1^{(k)}, v_2^{(k)}, \dots, v_6^{(k)}]$$

The first component, $v_1^{(k)}$, always represents the left-hand endpoint of an interval; $v_2^{(k)}$ is half the length of this interval; $v_3^{(k)}$, $v_4^{(k)}$, and $v_5^{(k)}$ are values of f at the left end, the middle point, and the right end of this interval, respectively; and $v_6^{(k)}$ is the value given by Simpson's rule for this interval.

The complete **Simpson's Adaptive Quadrature Algorithm** follows:

```

input  $a, b, \varepsilon, n$ 
 $\Delta \leftarrow b - a$ ;  $\Sigma \leftarrow 0$ ;  $h \leftarrow \Delta/2$ ;  $c \leftarrow (a + b)/2$ ;  $k \leftarrow 1$ 
 $\bar{a} \leftarrow f(a)$ ;  $\bar{b} \leftarrow f(b)$ ;  $\bar{c} \leftarrow f(c)$ 
 $S \leftarrow (\bar{a} + 4\bar{c} + \bar{b})h/3$ 
 $v^{(1)} \leftarrow [a, h, \bar{a}, \bar{c}, \bar{b}, S]$ 
while  $1 \leq k \leq n$ 
     $h \leftarrow v_2^{(k)}/2$ 
     $\bar{y} \leftarrow f(v_1^{(k)} + h)$ 
     $S^* \leftarrow (v_3^{(k)} + 4\bar{y} + v_5^{(k)})h/3$ 
     $\bar{z} \leftarrow f(v_1^{(k)} + 3h)$ 
     $S^{**} \leftarrow (v_4^{(k)} + 4\bar{z} + v_6^{(k)})h/3$ 
    if  $|S^* + S^{**} - v_6^{(k)}| < 30\varepsilon h/\Delta$  then
         $\Sigma \leftarrow \Sigma + S^* + S^{**} + [S^* + S^{**} - v_6^{(k)}]/15$ 
         $k \leftarrow k - 1$ 
        if  $k \leq 0$  then output  $\Sigma$ ; exit
    else
        if  $k \geq n$  then output failure; exit
         $\bar{v} \leftarrow v_5^{(k)}$ 
         $v^{(k)} \leftarrow [v_1^{(k)}, h, v_3^{(k)}, \bar{y}, v_4^{(k)}, S^*]$ 
         $k \leftarrow k + 1$ 
         $v^{(k)} \leftarrow [v_1^{(k-1)} + 2h, h, v_5^{(k-1)}, \bar{z}, \bar{v}, S^{**}]$ 
    end if
end do

```

In some programming languages, it is important to store the most frequently accessed quantities in the columns of arrays, which correspond to adjacent memory locations. If this is the case, then the workstack vector could be stored in a two-dimensional array $V(I, K) \leftarrow v_i^{(k)}$.

Here we present the adaptive quadrature algorithm with an explicit stack, but it is essentially a recursive procedure. First, try a simple Simpson's rule. If the error is satisfactory, accept the answer; otherwise, split the interval in half and recursively call the procedure on each half. Using recursion, we see the concept of the algorithm more easily. Moreover, the recursive algorithm is simple to code because most programming languages support recursion. We leave this as Computer Problem 7.5.1 (p. 513).

PROBLEMS 7.5

1. The local error criterion (Equation (3)) was used to obtain an *absolute* error bound. Try to establish a local error criterion that leads to a *relative* error bound

$$\left| \int_a^b f(x) dx - \sum_{i=1}^n S_i \right| \leq \left| \int_a^b f(x) dx \right| \varepsilon$$

2. Establish an approximation similar to Equation (7) using the trapezoid rule in the form

$$\int_u^v f(x) dv = T(u, v) - \frac{1}{12}(v-u)^3 f''(\xi)$$

where

$$T(u, v) = \frac{1}{2}(v-u)[f(u) + f(v)]$$

3. The composite trapezoid rule can be written as

$$I = T_m - \frac{1}{12}(v-u)h^2 f''(\xi)$$

where

$$I = \int_u^v f(x) dx$$

$$T_m = \frac{h}{2} \left[f(u) + 2 \sum_{i=1}^{m-1} f(u+ih) + f(v) \right]$$

Show how T_m and T_{2m} can be combined to obtain a better approximation to I . Here m is the number of subintervals in the interval $[u, v]$.

4. The composite Simpson's rule can be written as

$$I = S_{2m} - \frac{1}{180}(v-u)h^4 f^{(4)}(\xi)$$

where

$$I = \int_u^v f(x) dx$$

$$S_{2m} = \frac{h}{3} \left[f(u) + 2 \sum_{i=2}^m f(x_{2i-2}) + 4 \sum_{i=1}^m f(x_{2i-1}) + f(v) \right]$$

and $h = (v-u)/(2m)$, and $x_i = u + ih$ ($0 \leq i \leq 2m$). Here there are $2m$ subintervals in the interval $[u, v]$. Show how S_{2m} and S_{4m} can be combined to obtain a better approximation to I .

5. In Section 7.4 (p. 503), the following relationship, between the trapezoid rule over $[u, v]$ with m subintervals and the trapezoid rule over $[u, v]$ with $2m$ subintervals, was derived:

$$T_{2m} = \frac{1}{2}T_m + h \sum_{i=1}^m f(x_{2i-1})$$

where $h = (v-u)/(2m)$ and $x_i = u + ih$ ($0 \leq i \leq 2m$). Can this be used to establish an adaptive scheme?

COMPUTER
PROBLEMS 7.5

1. Program the adaptive algorithm and test it on these integrals:

- a. $\int_0^1 x^{1/2} dx$
- b. $\int_0^1 (1-x)^{1/2} dx$
- c. $\int_0^1 (1-x)^{1/4} dx$

2. Write and test the adaptive algorithm as a recursive procedure.

7.6 Sard's Theory of Approximating Functionals

A **linear functional** on a linear space is a linear map of that space into the scalar field (in this book, usually $\mathbb{R}$). For example, if the linear space is $C[a, b]$, an important linear functional φ is defined by

$$\varphi(f) = \int_a^b f(x) dx \qquad f \in C[a, b]$$

In numerical practice, the most basic functionals are **point evaluations**. Fixing x in $[a, b]$, we define a linear functional $\hat{x}$ by the equation

$$\hat{x}(f) = f(x) \qquad f \in C[a, b]$$

By taking linear combinations of point evaluations, we obtain more versatile functionals of the form ψ , defined by

$$\psi = \sum_{i=0}^n c_i \hat{x}_i \qquad \text{that is} \qquad \psi(f) = \sum_{i=0}^n c_i f(x_i) \qquad (1)$$

One can argue persuasively that in numerical work this is the most general type of functional that can be computed directly; other functionals (such as integration) must

be replaced by approximating functionals like ψ . This is precisely what is done in the subject of numerical integration and numerical differentiation.

A coherent theory of approximating functionals was developed by Arthur Sard in the years 1940–1970. This theory involves natural splines in an interesting manner. The type of functional that we shall approximate is given by the equation

$$\varphi(f) = \sum_{i=0}^N \left\{ \int_a^b \alpha_i(x) f^{(i)}(x) dx + \sum_{j=1}^n \beta_{ij} f^{(i)}(z_{ij}) \right\} \tag{2}$$

The points z_{ij} lie in $[a, b]$, and the functions α_i are assumed to be piecewise continuous on $[a, b]$. The functional φ can be applied to any $f \in C^N[a, b]$.

We shall say that a functional φ **annihilates** a space W if $\varphi(f) = 0$ for each $f \in W$. The **Peano kernels** of a functional φ such as in Equation (2) are the functions

$$K_m(t) = \frac{1}{m!} \varphi_x [(x - t)_+^m]$$

where $m \geq N$ and

$$(x - t)_+^m = \begin{cases} (x - t)^m & x \geq t \\ 0 & x < t \end{cases}$$

and the notation φ_x indicates that the functional applies to $(x - t)_+^m$ as a function of x . The function x_+^m (obtained when $t = 0$) is called a **truncated power function**.

EXAMPLE 1 Consider the functional φ defined by the equation

$$\varphi(f) = \int_0^\pi (\cos x) f'(x) dx$$

This illustrates Equation (2), with $N = 1$ and $\alpha_1(x) = \cos x$. What is the Peano kernel, K_1 , for φ ?

Solution The definition of the Peano kernels can be used to compute them explicitly. Notice that

$$\frac{d}{dx} (x - t)_+^m = m(x - t)_+^{m-1} \quad (m \geq 1)$$

Consequently, in this example,

$$\begin{aligned} K_1(t) &= \varphi_x [(x - t)_+^1] = \int_0^\pi (\cos x) \frac{d}{dx} (x - t)_+ dx \\ &= \int_0^\pi (\cos x) (x - t)_+^0 dx \\ &= \int_t^\pi (\cos x) dx = -\sin t \end{aligned}$$

■

The following result was proved in 1905. (In it we use Π_m as the space of all polynomials of degree at most m .)

THEOREM 1 Peano Kernel Theorem

If the functional in Equation (2) annihilates Π_m , then for all $f \in C^{m+1}[a, b]$,

$$\varphi(f) = \int_a^b K_m(t) f^{(m+1)}(t) dt \quad (3)$$

where $m \geq N$ and $K_m(t)$ is the Peano kernel of φ .

Proof By Taylor's Theorem with integral remainder (Theorem 5 in Section 1.1, p. 9),

$$f(x) = \sum_{k=0}^m \frac{1}{k!} f^{(k)}(a)(x-a)^k + r(x)$$

$$r(x) = \frac{1}{m!} \int_a^x f^{(m+1)}(t) (x-t)^m dt$$

Because φ annihilates Π_m , $\varphi(f) = \varphi(r)$. Now express r in the form

$$r(x) = \frac{1}{m!} \int_a^b f^{(m+1)}(t) (x-t)_+^m dt$$

From this, it follows that

$$\varphi(r) = \frac{1}{m!} \int_a^b f^{(m+1)}(t) \varphi_x[(x-t)_+^m] dt$$

This step involves passing the functional φ under the integral sign, which is justified by certain theorems in calculus that require the hypotheses we adopted for φ . ■

EXAMPLE 2 An illustration given by Sard [1963, p. 31] is the functional

$$\varphi(f) = \int_0^1 f(x) x^{-1/2} dx$$

Find an approximate formula

$$\psi(f) = c_1 f(0) + c_2 f(1)$$

that is an exact substitute for φ on the polynomial space Π_1 , and find its error.

Solution We require that $\varphi - \psi$ annihilate Π_1 . Using the method of undetermined coefficients (as in Section 7.2, p. 482), we first take $f(x) = 1$, getting

$$0 = \varphi(f) - \psi(f) = \int_0^1 x^{-1/2} dx - (c_1 + c_2) = 2 - c_1 - c_2$$

Taking $f(x) = x$, we obtain

$$0 = \varphi(f) - \psi(f) = \int_0^1 x^{1/2} dx - c_2 = \frac{2}{3} - c_2$$

Hence, $c_2 = 2/3$ and $c_1 = 4/3$. The Peano kernel K_1 for the functional $\varphi - \psi$ is

$$\begin{aligned} (\varphi_x - \psi_x)(x-t)_+^1 &= \int_0^1 (x-t)_+ x^{-1/2} dx - \frac{4}{3}(0-t)_+ - \frac{2}{3}(1-t)_+ \\ &= \int_t^1 (x-t)x^{-1/2} dx - \frac{2}{3}(1-t) \\ &= \frac{4}{3}t(t^{1/2} - 1) \end{aligned}$$

The Peano Kernel Theorem, applied to $\varphi - \psi$, now tells us that

$$\int_0^1 f(x)x^{-1/2} dx - \left[\frac{4}{3}f(0) + \frac{2}{3}f(1) \right] = \int_0^1 \frac{4}{3}t(t^{1/2} - 1)f''(t) dt$$

The right-hand side of this equation is an exact representation for the error when $f \in C^2[0, 1]$. Since the kernel is nonpositive on the interval $[0, 1]$, the Mean-Value Theorem for Integrals (Section 1.1, p. 9) gives

$$\int_0^1 \frac{4}{3}t(t^{1/2} - 1)f''(t) dt = f''(\xi) \int_0^1 \frac{4}{3}t(t^{1/2} - 1) dt = -\frac{2}{15}f''(\xi) \quad \blacksquare$$

If φ and ψ are two functionals that agree on Π_m , then their difference will annihilate Π_m . The difference will have a Peano kernel, K_m , and by the **Cauchy-Schwarz inequality**,

$$|\varphi(f) - \psi(f)| \leq \|K_m\|_2 \|f^{(m+1)}\|_2 \quad (4)$$

The norm with subscript 2 is the usual L_2 -norm on $[a, b]$. If there are parameters in ψ that are not fully determined by the requirement that $\varphi - \psi$ annihilate Π_m , then these parameters can be chosen to make the factor $\int_a^b K_m(t)^2 dt$ a minimum. In this way, we obtain a functional ψ that is a best approximation to φ in the sense of Sard. Schoenberg discovered that such *best* formulas can be obtained in a rather simple way, and this important theorem is presented next. In the theorem, $\varphi \circ L$ is the composition of φ and L , so that $(\varphi \circ L)(f) = \varphi(Lf)$.

THEOREM 2 Schoenberg Theorem

Let φ be the linear functional given in Equation (2). Let knots be prescribed thus: $a = t_0 < t_1 < \cdots < t_n = b$, with $n > N$. Among all the functionals of the form $\sum_{i=0}^n c_i \widehat{t}_i$ that agree with φ on Π_m , the best approximation to φ in the sense of Sard is $\varphi \circ L$, where L is the linear operator producing natural spline interpolants of degree $2m + 1$ at the given knots.

Proof Let ψ be of the form $\psi = \sum_{i=0}^n c_i \widehat{t_i}$, and assume that for any $p \in \Pi_m$, $\psi(p) = \varphi(p)$. Then $\varphi - \psi$ annihilates Π_m . Let K_m be its Peano kernel. Now $Lf = f$ if f is a natural spline of degree $2m + 1$ on the given knots. Because polynomials of degree $\leq m$ also are such natural splines, $Lp = p$ for $p \in \Pi_m$. Thus, $\varphi - \varphi \circ L$ also annihilates Π_m . Let $\overline{K}_m$ be its Peano kernel. We can show that

$$\int_a^b [\overline{K}_m(t)]^2 dt \leq \int_a^b [K_m(t)]^2 dt \quad (5)$$

The Peano kernel of the functional

$$\theta = \varphi \circ L - \psi = (\varphi - \psi) - (\varphi - \varphi \circ L)$$

is $\overline{\overline{K}}_m = K_m - \overline{K}_m$ and is given by

$$\overline{\overline{K}}_m(t) = \frac{1}{m!} \theta_x [(x - t)_+^m] \quad (6)$$

If $\{s_0, s_1, \dots, s_n\}$ is a cardinal basis for our natural spline space, then $s_i(t_j) = \delta_{ij}$ and L will have the form

$$Lf = \sum_{i=0}^n f(t_i) s_i$$

It follows that θ has the form

$$\theta(f) = \varphi(Lf) - \psi(f) = \sum_{i=0}^n f(t_i) \varphi(s_i) - \sum_{i=0}^n c_i f(t_i) = \sum_{i=0}^n \gamma_i f(t_i)$$

Thus, for $\overline{\overline{K}}_m$ we have the equation

$$\overline{\overline{K}}_m(t) = \frac{1}{m!} \sum_{i=0}^n \gamma_i (t_i - t)_+^m \quad (7)$$

Select a function g such that $g^{(m+1)} = \overline{\overline{K}}_m$. Then $g^{(2m+1)} = \overline{\overline{K}}_m^{(m)}$, and this is a step function or spline of degree 0. Consequently, g itself is a spline function of degree $2m + 1$, with knots $t_0, t_1, \dots, t_n$. In fact, g is a natural spline. To verify this, notice that by Equation (7), $\overline{\overline{K}}_m(t) = 0$ when $t \geq b$. Hence, $g^{(m+1)}(t) = 0$ for $t \geq b$. For $t \leq a \leq x$, we have from Equation (6):

$$\overline{\overline{K}}_m(t) = \frac{1}{m!} \theta_x [(x - t)^m]$$

Because θ annihilates Π_m , $\overline{\overline{K}}_m(t) = g^{(m+1)}(t) = 0$ when $-\infty < t \leq a$. Since g is a natural spline, $Lg = g$. Hence,

$$\int_a^b \overline{K}_m \overline{\overline{K}}_m dt = \int_a^b \overline{K}_m g^{(m+1)} dt = (\varphi - \varphi \circ L)(g) = 0$$

Equation (5) can now be established by writing

$$\begin{aligned}\int_a^b K_m^2 dt &= \int_a^b (\overline{K}_m + \overline{\overline{K}}_m)^2 dt = \int_a^b (\overline{K}_m^2 + 2\overline{K}_m\overline{\overline{K}}_m + \overline{\overline{K}}_m^2) dt \\ &= \int_a^b \overline{K}_m^2 dt + \int_a^b \overline{\overline{K}}_m^2 dt \geq \int_a^b \overline{K}^2 dt\end{aligned}$$

In terms of the inner product for functions, these calculations show that we have $\langle \overline{K}_m, \overline{\overline{K}}_m \rangle = 0$, whence by the Pythagorean Law,

$$\|K_m\|_2^2 = \|\overline{K}_m + \overline{\overline{K}}_m\|_2^2 = \|\overline{K}_m\|_2^2 + \|\overline{\overline{K}}_m\|_2^2 \geq \|\overline{K}_m\|_2^2 \quad \blacksquare$$

EXAMPLE 3 Let

$$\begin{aligned}\varphi(f) &= \int_{-1}^1 f(x) dx \\ \psi(f) &= c_1 f(-1) + c_2 f(0) + c_3 f(1)\end{aligned}$$

Considering only functionals ψ that agree with φ on Π_1 , find the best approximation to φ in the sense of Sard.

Solution We require the natural spline interpolation operator L for this situation. One can verify that

$$(Lf)(x) = a_0 + a_1 x + b_0(x+1)_+^3 + b_1(x)_+^3 + b_2(x-1)_+^3$$

where

$$\begin{aligned}a_0 &= [-f(-1) + 6f(0) - f(1)]/4 \\ a_1 &= [-5f(-1) + 6f(0) - f(1)]/4 \\ b_0 &= b_2 = -b_1/2 = [f(-1) - 2f(0) + f(1)]/4\end{aligned}$$

Then the best formula is given by

$$\begin{aligned}\psi(f) &= \varphi(Lf) = \int_{-1}^1 (Lf)(x) dx \\ &= 2a_0 + 4b_0 + \frac{1}{4}b_1 \\ &= \frac{3}{8}f(-1) + \frac{5}{4}f(0) + \frac{3}{8}f(1)\end{aligned} \quad \blacksquare$$

PROBLEMS 7.6

1. We seek an approximate formula

$$f'(x) \approx c_0 f(-1) + c_1 f(0) + c_2 f(1)$$

where x is a prescribed point in $(-1, 1)$. Find the conditions on the coefficients that are necessary and sufficient for the formula to be correct on Π_1 . Find the best approximation in the sense of Sard. Compute $\int_{-1}^1 K_1(t)^2 dt$ when the minimum is attained.

2. For Example 3, find the best constant in the error formula

$$v|\varphi(f) - \psi(f)| \leq c \|f''\|_2$$

3. Find the interpolating function $f(f(t_i) = \lambda_i, \quad 0 \leq i \leq n)$ that minimizes the integral $\int_0^n (f'')^2 dt$, assuming f'' is *piecewise* continuous.

4. Use the Peano Kernel Theorem on the functional

$$\varphi(f) = f(x+h) - f(x) - (h/2)[3f'(x) - f'(x-h)]$$

(x, h fixed), and prove that $\varphi(f) = (5/12)h^3 f'''(\xi)$, with $x-h < \xi < x+h$.

5. Use the Peano Kernel Theorem to obtain this result for Simpson's rule:

$$\int_0^2 f(x) dx = \frac{1}{3}[f(0) + 4f(1) + f(2)] - \frac{1}{90} f^{(4)}(\xi)$$

6. Use the Schoenberg Theorem to prove that

$$\left| f(2) + \frac{1}{4}f(0) - \frac{7}{8}f(1) - \frac{3}{8}f(3) \right| \leq \sqrt{\frac{5}{48}} \|f''\|_2$$

7.7 Bernoulli Polynomials and the Euler-Maclaurin Formula

The Romberg integration algorithm depends for its validity on the Euler-Maclaurin formula, Equation (6) in Section 7.4 (p. 504). We derive this formula here, after establishing some properties of the Bernoulli polynomials.

Bernoulli Polynomials

The Bernoulli polynomials form an infinite sequence with many useful properties. The first few of these polynomials are

$$\begin{aligned} B_0(t) &= 1 \\ B_1(t) &= t - \frac{1}{2} \\ B_2(t) &= t^2 - t + \frac{1}{6} \\ B_3(t) &= t^3 - \frac{3}{2}t^2 + \frac{1}{2}t \end{aligned}$$

They are defined by the equation

$$\sum_{k=0}^n \binom{n+1}{k} B_k(t) = (n+1)t^n \tag{1}$$

Recall that

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}$$

For example, if $n = 0$, then Equation (1) states that $B_0(t) = 1$. If $n = 1$, the equation states that $B_0(t) + 2B_1(t) = 2t$, from which we obtain $B_1(t) = t - \frac{1}{2}$. Since the coefficient of B_n in Equation (1) is $n + 1$, we can always solve for B_n in terms of $B_0, B_1, \dots, B_{n-1}$. The following theorem enumerates the properties of the Bernoulli polynomials that we shall require.

THEOREM 1 Theorem on Properties of Bernoulli Polynomials

The Bernoulli polynomials have these properties:

1. $B'_n = nB_{n-1} \quad (n \geq 1)$
2. $B_n(t + 1) - B_n(t) = nt^{n-1} \quad (n \geq 2)$
3. $B_n(t) = \sum_{k=0}^n \binom{n}{k} B_k(0)t^{n-k}$
4. $B_n(1 - t) = (-1)^n B_n(t)$

Proof Property 1 will be proved by induction. The case $n = 1$ is verified from the fact that $B_1(t) = t - \frac{1}{2}$ and $B_0(t) = 1$. Now suppose that $B'_k = kB_{k-1}$ for $k = 1, 2, \dots, n - 1$. Then by differentiating in Equation (1) and using Equation (1) again, we obtain

$$\sum_{k=1}^n \binom{n+1}{k} B'_k(t) = n(n+1)t^{n-1} = (n+1) \sum_{k=0}^{n-1} \binom{n}{k} B_k(t) \tag{2}$$

By means of the induction hypothesis, Equation (2) can be put into the form

$$(n+1)B'_n + \sum_{k=1}^{n-1} \binom{n+1}{k} kB_{k-1} = (n+1) \sum_{k=0}^{n-1} \binom{n}{k} B_k$$

Now divide by $n + 1$ and use the identity

$$\frac{k}{n+1} \binom{n+1}{k} = \binom{n}{k-1}$$

to obtain

$$B'_n + \sum_{k=1}^{n-1} \binom{n}{k-1} B_{k-1} = \sum_{k=0}^{n-1} \binom{n}{k} B_k$$

The cancellation of similar terms here leaves Property 1, which was to be proved. To prove Property 2, use Property 1 repeatedly to obtain

$$\begin{aligned} B''_n(t) &= n(n-1)B_{n-2}(t) \\ B'''_n(t) &= n(n-1)(n-2)B_{n-3}(t) \\ &\vdots \\ B_n^{(k)}(t) &= n(n-1) \cdots (n-k+1)B_{n-k}(t) \end{aligned}$$

Thus, the Taylor series for B_n is

$$B_n(t+h) = \sum_{k=0}^n \frac{1}{k!} B_n^{(k)}(t) h^k = \sum_{k=0}^n \binom{n}{k} B_{n-k}(t) h^k \quad (3)$$

Now set $h = 1$. Use the identity

$$\binom{n}{k} = \binom{n}{n-k}$$

and Equation (1) to obtain Property 2:

$$\begin{aligned} B_n(t+1) &= \sum_{k=0}^n \binom{n}{k} B_k(t) = B_n(t) + \sum_{k=0}^{n-1} \binom{n}{k} B_k(t) \\ &= B_n(t) + nt^{n-1} \end{aligned}$$

We obtain Property 3 by putting $t = 0$ and $h = t$ in Equation (3).

To derive Property 4, use Property 2 with t replaced by $-t$. The result is

$$\begin{aligned} B_n(1-t) - B_n(-t) &= n(-1)^{n-1} t^{n-1} \\ &= (-1)^{n-1} [B_n(t+1) - B_n(t)] \end{aligned}$$

When this is written in the form

$$(-1)^n B_n(t+1) - B_n(-t) = (-1)^n B_n(t) - B_n(1-t) \equiv F(t)$$

we recognize it as an equation of the type $F(t+1) = F(t)$, which implies that F is periodic with period 1. Because F is a polynomial, it must be constant. Hence,

$$(-1)^n B_n(t) - B_n(1-t) = c_n \quad (4)$$

Now differentiate in Equation (4) and use Property 1:

$$(-1)^n B'_n(t) + B'_n(1-t) = (-1)^n n B_{n-1}(t) + n B_{n-1}(1-t) = 0$$

This is equivalent to Property 4. ■

LEMMA 1 Lemma on Bernoulli Polynomials

The function $G(t) = B_{2n}(t) - B_{2n}(0)$ has no zeros in the open interval $(0, 1)$.

Proof We begin by setting $t = 0$ in Properties 2 and 4 of Theorem 1. The result is

$$B_n(0) = B_n(1) = (-1)^n B_n(0)$$

Consequently, $B_3(0) = B_5(0) = B_7(0) = \cdots = 0$. Now suppose that G has a zero in $(0, 1)$. Since $G(0) = G(1) = 0$, we see by Rolle's Theorem that G' has two zeros in $(0, 1)$. Since B_{2n-1} is a multiple of G' , B_{2n-1} has two zeros in $(0, 1)$. But B_{2n-1} vanishes also at 0 and 1, so B'_{2n-1} has three zeros in $(0, 1)$. Consequently, B_{2n-2} has three zeros in $(0, 1)$. Continuing in this way, we conclude that B_k , for odd indices $k < 2n$, must have at least two zeros in $(0, 1)$. Hence, B_3 must have two zeros in $(0, 1)$ as well as the two zeros at 0 and 1. This is manifestly impossible because B_3 is a cubic polynomial. ■

THEOREM 2 Euler-Maclaurin Formula

If the function f possesses $2n$ continuous derivatives in $[0, 1]$, then

$$\int_0^1 f(t) dt = \frac{1}{2}[f(0) + f(1)] - \sum_{k=0}^{n-1} \frac{b_{2k}}{(2k)!} [f^{(2k-1)}(1) - f^{(2k-1)}(0)] + R$$

where

$$\begin{cases} b_k = B_k(0) \\ R = -\frac{b_{2n}}{(2n)!} f^{(2n)}(\xi) \end{cases} \quad (0 < \xi < 1)$$

Proof From Property 1 of Theorem 1, we have

$$B_n(t) = \frac{1}{n+1} B'_{n+1}(t)$$

Using this formula and integration by parts, we write

$$\int_0^1 f(t) dt = \int_0^1 f(t) B_0(t) dt = B_1(t) f(t) \Big|_0^1 - \int_0^1 B_1(t) f'(t) dt \quad (5)$$

Since $B_1(1) = \frac{1}{2}$ and $B_1(0) = -\frac{1}{2}$, Equation (5) can be written in the form

$$\int_0^1 f(t) dt = \frac{1}{2}[f(1) + f(0)] - \int_0^1 B_1(t) f'(t) dt$$

The integration in this equation also can be carried out with an integration by parts, and the result of doing so is

$$\int_0^1 f(t) dt = \frac{1}{2}[f(1) + f(0)] - \frac{b_2}{2}[f'(1) - f'(0)] + \frac{1}{2} \int_0^1 B_2(t) f''(t) dt$$

This process can be continued. The formulas

$$\begin{aligned} B_n(0) &= B_n(1) = b_n \\ b_3 &= b_5 = b_7 = \cdots = 0 \end{aligned}$$

previously mentioned are helpful. After $2n$ steps, we have

$$\begin{aligned} \int_0^1 f(t) dt &= \frac{1}{2}[f(1) + f(0)] - \sum_{k=1}^n \frac{b_{2k}}{(2k)!} [f^{(2k-1)}(1) - f^{(2k-1)}(0)] \\ &\quad + \frac{1}{(2n)!} \int_0^1 B_{2n}(t) f^{(2n)}(t) dt \end{aligned}$$

The last term in the sum can be expressed in the form

$$\frac{b_{2n}}{(2n)!} [f^{(2n-1)}(1) - f^{(2n-1)}(0)] = \frac{b_{2n}}{(2n)!} \int_0^1 f^{(2n)}(t) dt$$

This permits the formula to be written as in the statement of the theorem, except that the remainder now is

$$R = \frac{1}{(2n)!} \int_0^1 [B_{2n}(t) - b_{2n}] f^{(2n)}(t) dt \tag{6}$$

By Lemma 1 on Bernoulli Polynomials, the function $B_{2n}(t) - b_{2n}$ does not change sign in $[0, 1]$. Hence, the Mean-Value Theorem for Integrals can be applied in Equation (6) to write R in the form

$$R = \frac{1}{(2n)!} f^{(2n)}(\xi) \int_0^1 [B_{2n}(t) - b_{2n}] dt$$

Because $B_{2n} = B'_{2n+1}/(2n + 1)$ and because $B_{2n+1}(t)$ is 0 at $t = 0$ and $t = 1$, we finally obtain

$$R = -\frac{b_{2n}}{(2n)!} f^{(2n)}(\xi) \quad \blacksquare$$

PROBLEMS 7.7

- 1. Prove that the leading term in $B_n(t)$ is t^n and that the term t^{n-1} has coefficient $-n/2$.
- 2. Use Property 2 in Theorem 1 to prove that
$$1^p + 2^p + \cdots + n^p = [B_{p+1}(n+1) - B_{p+1}(0)]/(p+1)$$
- 3. Show that the functions $B_n(x) - B_n(0)$ have a simple zero at $\frac{1}{2}$ when n is odd.
- 4. Show that the numbers $B_0(0), B_2(0), B_4(0), \dots$ alternate in sign.
- 5. Prove that for even n , B_n has at least two roots in $(0, 1)$, and for odd n , B_n has at least one root in $(0, 1)$.
- 6. Prove that the binomial coefficients $\binom{n}{m}$ are integers. *Hint:* You could first prove Pascal's Law:

$$\binom{n}{m-1} + \binom{n}{m} = \binom{n+1}{m}$$